
M²NuFFT—A Computationally Efficient Suboptimal Power Spectrum Estimator for Fast Exploration of Nonuniformly Sampled Time Series

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Abstract

Nonuniformly sampled signals are prevalent in real-world applications. However, estimating their power spectra from finite samples poses a significant challenge. The optimal solution—Bronez Generalized Prolate Spheroidal Sequence (GPSS) by solving the associated Generalized Eigenvalue Problem (GEP)—is computationally intensive and thus impractical for large datasets. This paper describes a fast, nonparametric method: Multiband-Multitaper Nonuniform Fast Fourier Transform (M²NuFFT), which substantially reduces computational burden while maintaining statistical efficiency. The algorithm partitions the signal frequency band into multiple sub-bands. Within each sub-band, optimal tapers are computed at a nominal analysis band and shifted to other analysis bands using the Nonuniform Fast Fourier Transform (NuFFT), avoiding repeated GEP computations. Spectral power within the analysis band is then estimated as the average power across the taper outputs. For the special case where the nominal band is centered at zero frequency, tapers can be approximated via cubic spline interpolation of Discrete Prolate Spheroidal Sequence (DPSS), eliminating GEP computation entirely. This reduces the complexity from $O(N^4)$ to as low as $O(N \log N + N \log(1/\epsilon))$. Statistical properties of the estimator, assessed using Bronez GPSS theory, reveal that the bias and variance bound of the M²NuFFT estimator are identical to those of the optimal estimator. Additionally, the degradation of bias bound indicates deviation from optimality. Finally, we propose an extension of Thomson F -test to test periodicity in nonuniform samples. The estimator's performance is validated through simulation and real-world data, demonstrating its practical applicability. The MATLAB code of the fast algorithm is available on GitHub [1].

Table 1. Abbreviations Used Frequently

Abbreviation	Description
M ² NuFFT	Multiband-Multitaper Nonuniform Fast Fourier Transform
BG	Bronez GPSS
DFT	Discrete Fourier Transform
DPSS	Discrete Prolate Spheroidal Sequence
GEP	Generalized Eigenvalue Problem
GPSS	Generalized Prolate Spheroidal Sequence
MDSS	Missing-Data Slepian Sequence
MTLS	Multitaper Lomb-Scargle periodogram
NuFFT	Nonuniform Fast Fourier Transform
PSD	Power Spectral Density

Table 2. Mathematical Symbols Used Frequently

Symbol	Description
E_i	Frequency shift operator
I	Number of analysis bands $\mathcal{A}_i$
$J_k(\mathcal{A}_i)$	k -th eigenvalue coefficient of analysis band $\mathcal{A}_i$
$J_k(f)$	k -th eigenvalue coefficient of uniformly sampled signal
K	Number of tapers
N	Sample size of signal $\mathbf{x}$
$P(\mathcal{A})$	Integrated spectrum (power) in analysis band $\mathcal{A}$
Q	Number of sub-signal bands $\mathcal{B}^q$ in signal band $\mathcal{B}$, for $q = 0, 1, \dots, Q - 1$
$S(f)$	Power spectral density of uniformly sampled signal
β	Sampling interval of missing data sampling scheme
$\hat{L}_k^i$	Estimate of taper suboptimality measure L_k^i
$\hat{P}(\mathcal{A}_i)$	An estimate of $P(\mathcal{A})$ on analysis band $\mathcal{A}_i$
λ_k^0	Equivalent to $\lambda_k(\mathcal{A}_0)$
λ_k^i	Equivalent to $\lambda_k(\mathcal{A}_i)$
$ J_k(\mathcal{A}_i) ^2$	k -th eigenspectrum of analysis band $\mathcal{A}_i$
$ J_k(f) ^2$	k -th eigenspectrum of uniformly sampled signal
$\mathbb{I}$	Identity matrix
ϵ	Computation precision of fast NuFFT algorithm
$\mathbf{R}(\mathcal{A}_0)$	GPSS matrix for analysis band $\mathcal{A}_0$
$\mathbf{R}(\mathcal{A}_i)$	GPSS matrix for analysis band $\mathcal{A}_i$, for $i = 0, 1, \dots, I - 1$
$\mathbf{R}(\mathcal{A} n, m)$	Element (n, m) of GPSS matrix $\mathbf{R}(\mathcal{A})$
$\mathbf{R}(\mathcal{B})$	GPSS matrix for signal band $\mathcal{B}$
$\mathbf{R}(\mathcal{B} n, m)$	Element (n, m) of GPSS matrix $\mathbf{R}(\mathcal{B})$
$\mathbf{w}_k^0$	Simplified notation of $\mathbf{w}_k(\mathcal{A}_0)$, if context is clear.
$\mathbf{w}_k^i$	Simplified notation of $\mathbf{w}_k(\mathcal{A}_i)$, if context is clear.
$\mathbf{x}$	Vector of a weakly stationary, band-limited Gaussian process
$\mathcal{A}_0$	Analysis band centered at f_{c_0} , or nominal band

Symbol	Description
$\mathcal{A}_i$	Analysis band centered at f_{c_i} , for $i = 1, 2, \dots, I - 1$
$\mathcal{A}$	Generic analysis band
$\mathcal{B}^q$	q -th sub-band of signal band $\mathcal{B}$, $q = 0, 1, \dots, Q - 1$. Simplified as $\mathcal{B}$, if context is clear.
$\mathcal{B}$	Entire signal band of signal $\mathbf{x}$
$\mathcal{E}_i$	Suboptimality measure at analysis band $\mathcal{A}_i$
$\hat{\mathbf{w}}_k^0$	Estimate of $\mathbf{w}_k^0$
f_c	Center frequency of generic analysis band $\mathcal{A}$
f_s	Sampling frequency
f_w	Bandwidth of an analysis band $\mathcal{A}$, whose frequency resolution is $2f_w$.
$f_{\max}$	Maximum frequency of signal band $\mathcal{B}$
f_{c_i}	Center frequency of analysis band $\mathcal{A}_i$, for $i = 0, 1, \dots, I - 1$
t_n	Sampling time of n -th sample, for $n = 1, 2, \dots, N$
$v_n^{(k)}(N, f_w)$	n -th element of the vector with portion of DPSS, order k
$x(n)$	Sample at time t_n of signal $\mathbf{x}$, equivalent to $x(t_n)$. Often refers to a uniformly sampled signal.
$x(t_n)$	Sample at time t_n of signal $\mathbf{x}$

1 Introduction

Power spectrum estimation is a fundamental tool in a wide variety of scientific and engineering disciplines [2, 3, 4, 5], including signal processing, communication, machine learning, physical science, and biomedical data analysis [6, 7]. It allows for the characterization of the second moments of a time series, elucidating periodicities, oscillatory behavior, and correlation structures in a signal process. These attributes are crucial to numerous applications.

Despite its extensive lineage [2], power spectrum estimation continues to be an active research domain. The primary challenge resides in estimating the spectrum in a way to minimize bias and ensure statistical robustness, often from a finite sample of the signal. In many instances, only a single realization (trial) of the underlying process is available, making the estimation problem inherently ill-posed [8, 9]. The continuous Power Spectral Density (PSD) cannot be directly observed and must be estimated from discrete, limited samples. This constraint introduces bias and variance into the estimation, primarily due to spectral leakage caused by time-domain windowing—equivalent to convolution with the window’s Fourier transform in the frequency domain. Traditional power spectrum estimators such as the periodogram [10] are computationally simple but suffer from high variance in performance. In addition, in many real-world applications, the signal is often nonuniformly¹ sampled. This includes scenarios such as network packet data transfer [11], laser Doppler anemometry [12, 13], geophysics [14], atomic clock analysis [15], astronomy [16, 17, 18, 19, 20, 21], computer tomography [22], genetics [23], biosensing optimization [24], and biological signals [25, 26, 27, 28]. Nonuniform sampling often leads to increased sidelobe leakage and inflated bias in spectral estimates [29].

This paper focuses on a nonparametric solution to power spectrum estimation problem, in contrast to parametric methods that assume a specific model of the time series [4]. Nonparametric methods are particularly suitable for rapid, exploratory analysis of large datasets, especially when the underlying model

¹Some other terminology has been interchangeably adopted in literature, such as unevenly, irregularly, and unequally sampled signal.

is unknown. In such cases, Thomson’s multitaper method proves to be a powerful tool [8]. This method employs the Discrete Prolate Spheroidal Sequence (DPSS), also known as Slepian sequence [30], denoted as $v_n^{(k)}(N, f_w)$, where k indexes the taper (with $1 \leq k \leq K$), N is the signal length, and f_w is the bandwidth. The method computes the Discrete Fourier Transform (DFT) of the uniformly sampled signal, $x(n)$, $1 \leq n \leq N$, weighted by the DPSS taper,

$$J_k(f) = \sum_{n=1}^N \left[x(n) v_n^{(k)}(N, f_w) \right] e^{-j2\pi f n / f_s}, \quad (1)$$

where f_s is the sampling frequency. The power spectrum estimate is then computed by averaging the squared magnitudes of these eigencoefficients $J_k(f)$, $|J_k(f)|^2$, known as the k -th eigenspectrum,

$$\hat{P}(f) = \frac{1}{K} \sum_{k=1}^K |J_k(f)|^2. \quad (2)$$

Adaptive weighting schemes for $|J_k(f)|^2$ ’s is available to further improve estimation quality [8, 4]. The multitaper approach achieves a principled tradeoff between resolution, bias, and variance, and has been extensively validated across diverse applications [31, 32, 4, 33, 34].

Extending the multitaper scheme to nonuniformly sampled signal is desirable. However, direct application of the classical multitaper approach is nontrivial, as estimator performance under nonuniform sampling depends on more than just frequency resolution. Lepage (2009) [35] proposed a direct generalization of Thomson’s original approach [8], replacing the DFT with the “irregular DFT (irDFT)” and subsequently replacing the Dirichlet-type kernel with a sampling scheme-dependent, Hermitian, Toeplitz kernel. This method demonstrated superior performance compared to competitive multitaper estimates computed from the uniformly sampled data using interpolation. Springford (2020) [18] adapted the Thomson multitaper method to enhance the estimation from the Lomb-Scargle (LS) periodogram [16, 17], a technique widely employed in astronomy. Dodson-Robinson and Haley (2024) [36] evaluated the performance of Chave’s Missing-Data Slepian Sequence (MDSS) [37] and further suggested the application of an F -test to assess periodicity in nonuniformly sampled data. Patil et al. [20, 21] provided compelling empirical evidence that combining interpolated DPSS with the Nonuniform Fast Fourier Transform (NuFFT) can significantly enhance spectral and harmonic analysis of astrophysical signals. Additionally, recent developments in compressive sensing offer alternative strategies for spectrum recovery from randomly sampled data [38], particularly when the signal exhibits frequency sparsity. While these approaches have merit in various aspects, a comprehensive evaluation of their statistical properties in terms of bias, variance, and optimality has not been adequately evaluated. Moreover, their computational efficiency has not been systematically addressed.

In contrast to heuristic approaches, the seminal work by Bronez (1985, 1988) [39, 40] proposed an optimal estimator based on the study of the first and second moments of quadratic spectral estimator (see Section 2) for arbitrary sampling times. This method calculates the optimal weight sequence for each analysis band, known as the Generalized Prolate Spheroidal Sequence (GPSS), by solving a Generalized Eigenvalue Problem (GEP). This work established an optimality criterion for the performance of power spectrum estimators in the general case of sampling schemes. However, since the number of analysis bands is in general proportional to N , and the GPSS has to be estimated for each analysis band, the computational cost is prohibitively high for large N .

In this study, we have developed a fast algorithm, termed Multiband-Multitaper Nonuniform Fast Fourier Transform (M^2 NuFFT), by integrating Thomson’s and Bronez’s multitaper estimators. Specifically, we partition the entire signal band, $\mathcal{B}$, into multiple sub-bands, $\mathcal{B}^q$, $q = 0, 1, \dots, Q - 1$, enabling parallel computing. Within each sub-band, the core idea is to estimate the optimal weight sequence $\mathbf{w}_k^0$ (i.e., the GPSS) on a nominal analysis band $\mathcal{A}_0$ (see Section 3). Rather than solving the computationally expensive GEP for

other analysis bands $\mathcal{A}_i$, for $1 \leq i \leq I - 1$, we efficiently shift $\mathbf{w}_k^0$ to the $\mathcal{A}_i$, using the NuFFT. This ensures that for each sub-signal band, we only need to solve GEP once. Consequently, the eigencoefficients $J_k(\mathcal{A}_i)$ and the integrated spectrum $\hat{P}(\mathcal{A}_i)$ are readily computed from K eigenspectra $|J_k(\mathcal{A}_i)|^2$ for each $\mathcal{A}_i$. For the special case where the nominal band $\mathcal{A}_0$ is centered at $f_{c0} = f_0 = 0$ Hz, further computational reduction may be achieved by interpolating the uniformly spaced $v_n^{(k)}(N, f_w)$ to the nonuniform grid t_n using cubic splines [18] to approximate the GPSS at $\mathcal{A}_0$, $\hat{\mathbf{w}}_k^0$. In this case, the overall computational complexity is comparable to that of fast NuFFT, approximately $O(N \log N + N \log(1/\epsilon))$, where ϵ is the precision of computations [41, 42].

We evaluated the statistical properties of the proposed method, focusing on bias, variance, and suboptimality, using the theory developed in Bronez GPSS [40]. Our findings indicate that the bias and variance bounds are consistent with those of the optimal method. Additionally, we suggest that the suboptimality of the fast algorithm may be quantified by the difference between the approximate and optimal eigenvalues. Furthermore, an F -test has been implemented to assess periodicity in nonuniformly sampled time series [8, 19, 21].

Importantly, we emphasize that although the main theoretical framework used in this study was first proposed in the 1980s and 1990s, it remains highly relevant today. Specifically, the multitaper method of spectral estimation for arbitrary sampling schemes [39, 40, 32, 43] continues to be a powerful tool for addressing this challenging problem. This is evidenced by its application in several recent studies [37, 15, 19, 20, 21, 24, 27], underscoring its ongoing importance and the necessity for a computationally efficient implementation.

The remainder of the paper is organized as follows. We provide an overview of the Bronez GPSS method in Section 2, serving as the theoretical basis for the following developments. In Section 3 we develop the fast M²NuFFT algorithm² and evaluate its statistic properties in the context of Bronez GPSS theory. Section 4 is dedicated to analyzing taper approximation errors through numerical experiments. The performance evaluation of the estimator, which includes both simulation results and a real-world application, is presented in Section 5. Section 6 offers a broader discussion, followed by concluding remarks in Section 7. The MATLAB (MathWorks, Natick, MA) code of the fast algorithm (M²NuFFT, Table 3) is publicly available on GitHub [1].

2 Overview of Bronez GPSS Optimal Approach

The Bronez GPSS (BG) is an extension of the quadratic spectral estimator, developed to analyze nonuniformly sampled processes [39, 40]. It is an optimal nonparametric method in the sense that it is unbiased in the context of white noise, and it minimizes the variance and bias bounds for a given frequency resolution.

Consider $x(t_n)$ a weakly stationary, band-limited Gaussian process, available on a set of arbitrary sampling points t_n , $1 \leq n \leq N$, where N is the total number of samples. Instead of directly estimating the PSD, $S(f)$, BG estimates the integrated spectrum (i.e. the power), $P(\mathcal{A}) = \int_{\mathcal{A}} S(f) df$, contained in an analysis band of interest $\mathcal{A} = \{f : |f - f_c| \leq f_w\}$, where f_c is the center frequency and f_w the bandwidth. Note that $2f_w$ is the desired frequency resolution. A complete spectral analysis involves estimating $P(\mathcal{A})$ for a set of analysis bands, $\mathcal{A}$, to cover the entire signal band $\mathcal{B} = \{f : |f| \leq f_{\max}\}$, where $f_{\max}$ is presumably the maximum frequency of the signal [40]. The estimator can be expressed as

$$\hat{P}(\mathcal{A}) = \frac{1}{K} \mathbf{x}^* \mathbf{Q}(\mathcal{A}) \mathbf{x}, \quad (3)$$

where $\mathbf{x} = [x(t_1), x(t_2), \dots, x(t_N)]'$, the prime, $'$, denotes vector transposition, and the asterisk, $*$, denotes complex conjugate transposition. The $N \times N$ positive semidefinite Hermitian weight matrix $\mathbf{Q}(\mathcal{A})$ depends

²Dr. G.M. Eadie of University of Toronto kindly noted (personal communication) that their group had already adopted a similar name (i.e. mtNUFFT) for their methods [20, 21]. We formulated the acronym and theory independently.

on the analysis band $\mathcal{A}$. Here, $K \leq N$ is the rank of $\hat{P}(\mathcal{A})$. The weight matrix $\mathbf{Q}(\mathcal{A})$ can be factorized as $\mathbf{Q}(\mathcal{A}) = \mathbf{\Psi}(\mathcal{A})\mathbf{\Psi}^*(\mathcal{A})$, where $\mathbf{\Psi}(\mathcal{A})$ is an $N \times K$ matrix. The power spectrum estimator is then given by

$$\hat{P}(\mathcal{A}) = \frac{1}{K} \sum_{k=1}^K |\mathbf{w}_k^*(\mathcal{A})\mathbf{x}|^2, \quad (4)$$

where $\mathbf{w}_k(\mathcal{A})$, $1 \leq k \leq K$, is the columns of $\mathbf{\Psi}(\mathcal{A})$.

Assuming that the number of weight sequences, also known as tapers, K , is predetermined, the optimal tapers $\mathbf{w}_k(\mathcal{A})$ are derived based on the constraints imposed on estimator bias, variance bound, and bias bound.

2.1 Bias Constraint

The estimator, as defined in (4), is constrained to be unbiased when the true spectral density is flat, e.g., $S(f) = 1$. Given the expectation of the estimate

$$E\{\hat{P}(\mathcal{A})\} = \int_{\mathcal{B}} \frac{S(f)}{K} \sum_{k=1}^K |W_k(f)|^2 df, \quad (5)$$

where $W_k(f)$ is the DFT of $\mathbf{w}_k(\mathcal{A}) = [w_k(t_1), w_k(t_2), \dots, w_k(t_n)]$,

$$W_k(f) \triangleq \sum_{n=1}^N w_k(t_n) e^{-j2\pi f t_n}, \quad (6)$$

to minimize the bias, $E\{\hat{P}(\mathcal{A})\} - P(\mathcal{A})$, the weight sequences, $\mathbf{w}_k(\mathcal{A})$, must satisfy

$$\int_{\mathcal{B}} \frac{1}{K} \sum_{k=1}^K |W_k(f)|^2 df = \int_{\mathcal{A}} df, \quad (7)$$

which is equivalent to

$$\frac{1}{K} \sum_{k=1}^K \mathbf{w}_k^*(\mathcal{A}) \mathbf{R}(\mathcal{B}) \mathbf{w}_k(\mathcal{A}) = 2f_w, \quad (8)$$

where $\mathbf{R}(\mathcal{B})$ is the GPSS matrix on signal band $\mathcal{B}$. It is an $N \times N$ positive definite Hermitian matrix, whose elements are

$$\begin{aligned} \mathbf{R}(\mathcal{B}|n, m) &= \int_{\mathcal{B}} e^{j2\pi f(t_n - t_m)} df \\ &= \frac{\sin[2\pi f_{\max}(t_n - t_m)]}{\pi(t_n - t_m)}. \end{aligned} \quad (9)$$

2.2 Variance Bound

For a Gaussian process, it can be shown (Appendix B and equation (16) in [40]) that the variance of the estimator can be bounded above by

$$\text{VAR}\{\hat{P}(\mathcal{A})\} \leq S_{\max}^2 \cdot \mathbf{V}\{\mathbf{w}_1(\mathcal{A}), \dots, \mathbf{w}_K(\mathcal{A})\},$$

where $S_{\max} = \sup_{f \in [B]} S(f)$ is the maximum value of the spectral density, and $\mathbf{V}\{\mathbf{w}_k(\mathcal{A})\}_{k=1}^K$ is the bound factor

$$\begin{aligned} & \mathbf{V}\{\mathbf{w}_1(\mathcal{A}), \dots, \mathbf{w}_K(\mathcal{A})\} \\ &= \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K \left| \int_{\mathcal{B}} W_k(f)^* W_l(f) df \right|^2 \\ &= \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K |\mathbf{w}_k^*(\mathcal{A}) \mathbf{R}(\mathcal{B}) \mathbf{w}_l(\mathcal{A})|^2. \end{aligned} \quad (10)$$

Choosing the weight sequence to minimize (10), while satisfying (8), leads to the sequence normalization requirement

$$\mathbf{w}_k^*(\mathcal{A}) \mathbf{R}(\mathcal{B}) \mathbf{w}_k(\mathcal{A}) = 2f_w \quad 1 \leq k \leq K. \quad (11)$$

2.3 Bias Bound

By considering only the broad band bias [8], the errors due to frequencies outside analysis band $\mathcal{A}$, an approximate bias of estimation can be bounded above by

$$\widehat{BIAS}\{\hat{P}(\mathcal{A})\} \leq S_{\max} \cdot \mathbf{B}\{\mathbf{w}(\mathcal{A})_1, \dots, \mathbf{w}(\mathcal{A})_k\}, \quad (12)$$

where $\mathbf{B}\{\mathbf{w}(\mathcal{A})\}_{k=1}^K$ is the bias bound factor (defined in equation (20b) in [40]). Given the normalization requirement (11), choosing $\mathbf{w}_k(\mathcal{A})$ to minimize the bound factor

$$\begin{aligned} & \mathbf{B}\{\mathbf{w}(\mathcal{A})_1, \dots, \mathbf{w}(\mathcal{A})_k\} \\ &= \int_{\mathcal{B}-\mathcal{A}} \frac{1}{K} \sum_{k=1}^K |W_k(f)|^2 df \\ &= \frac{1}{K} \sum_{k=1}^K \mathbf{w}_k^*(\mathcal{A}) [\mathbf{R}(\mathcal{B}) - \mathbf{R}(\mathcal{A})] \mathbf{w}_k(\mathcal{A}), \end{aligned} \quad (13)$$

where the cut integral is defined as $\int_{\mathcal{B}-\mathcal{A}} = \int_{\mathcal{B}} - \int_{\mathcal{A}}$. This results in the GEP

$$\mathbf{R}(\mathcal{A}) \mathbf{w}_k(\mathcal{A}) = \lambda_k^{\mathcal{A}} \mathbf{R}(\mathcal{B}) \mathbf{w}_k(\mathcal{A}), \quad 1 \leq k \leq K, \quad (14)$$

where $\mathbf{R}(\mathcal{B})$ is the GPSS matrix on the signal band shown in (9) and $\mathbf{R}(\mathcal{A})$ is the GPSS matrix on the analysis band

$$\begin{aligned} \mathbf{R}(\mathcal{A}|n, m) &= \int_{\mathcal{A}} e^{j2\pi f(t_n - t_m)} df \\ &= \frac{\sin[2\pi f_w(t_n - t_m)]}{\pi(t_n - t_m)} e^{j2\pi f_c(t_n - t_m)}. \end{aligned} \quad (15)$$

The GEP (14) has N independent solutions $\{\lambda_k^{\mathcal{A}}, \mathbf{w}_k(\mathcal{A})\}$, $1 \leq k \leq N$, for the analysis band $\mathcal{A}$. The weight sequences corresponding to the K largest eigenvalues are chosen to minimize the bound factors.

The computation of the GEP (14) requires $O(N^3)$ operations for each analysis band of interest. In general, the number of bands is proportional to the number of samples, N , and thus makes the total computational load on the order of $O(N^4)$. The computational demand may be impractical when N is large.

2.4 Analysis in a Sub-Signal Band

In the preceding discussion, the signal band $\mathcal{B}$ was assumed to span the entire frequency range of the signal, which is typically the case in practice. However, as discussed in Section 3, it can be advantageous to consider sub-signal bands for localized analysis and parallel computation. A sub-band is defined as $\mathcal{B}^q = \{f : f_{\min}^q \leq |f| \leq f_{\max}^q\}$, where $f_{\min}^q \geq 0$ and $f_{\max}^q \leq f_{\max}$ denote the lower and upper bounds of $\mathcal{B}^q$, and $q = 0, 1, \dots, Q-1$ indexes the Q disjoint sub-bands, collectively covering the entire signal band, $\mathcal{B} = \bigcup_{q=0}^{Q-1} \mathcal{B}^q$. In this paper, we assume that the partition of the signal band into Q sub-band $\mathcal{B}^q$ is predetermined (Table 3, but see the discussion in Section 6).

From the GPSS formulation in equation (9) and the Bronez's theorem stated in [40], the GPSS matrix $\mathbf{R}(\mathcal{B}^q)$ corresponding to each sub-band $\mathcal{B}^q$ remains a positive definite Hermitian matrix (A), whose elements are

$$\begin{aligned} \mathbf{R}(\mathcal{B}^q|n, m) &= \int_{-f_{\max}^q}^{f_{\max}^q} e^{j2\pi f(t_n - t_m)} df - \int_{-f_{\min}^q}^{f_{\min}^q} e^{j2\pi f(t_n - t_m)} df \\ &= 2 \frac{\cos[\pi(f_{\max}^q + f_{\min}^q)(t_n - t_m)] \sin[\pi(f_{\max}^q - f_{\min}^q)(t_n - t_m)]}{\pi(t_n - t_m)}, \end{aligned} \quad (16)$$

where $f_{\min}^q < f_{\max}^q$. The GEP in equation (14) can be solved independently for each $\mathcal{B}^q$ to obtain the optimal weight sequences $\mathbf{w}_k(\mathcal{A}^q)$. Importantly, the constraints governing bias control, normalization, and optimality—namely, equations (8), (11) and (14) extend naturally to each sub-band $\mathcal{B}^q$. The full frequency band $\mathcal{B}$ can thus be viewed as a special case, where $f_{\max}^q = f_{\max}$, $f_{\min}^q = 0$, and $Q = 1$.

3 Multiband-Multitaper Nonuniform Fast Fourier Transform

In this section, we present the core structure of the Multiband-Multitaper Nonuniform Fast Fourier Transform (M²NuFFT) method, designed for rapid power spectrum estimation in nonuniformly sampled time series. The derivation of this method assumes that the series follows a weakly-stationary, band-limited Gaussian process, similar to previously introduced methods [39, 40]. The number of weight sequences, or tapers, denoted as K , is predetermined and correlates with the properties of the tapers obtained. We will evaluate the statistical performance of the estimator based on bias measure, variance bound, and sidelobe leakage. The quantification of leakage may serve as a measure of suboptimality.

3.1 Multiband Partition of Signal Band

To reduce computational load and enable parallelization, we partition the full signal band $\mathcal{B}$ into Q non-overlapping sub-bands, $\mathcal{B}^q$, $q = 0, 1, \dots, Q-1$. Each sub-band contains a group of analysis bands (c.f., Section 2.4). This partition is particularly effective when $Q \ll N$, as it requires the GEP be solved only once per sub-band, significantly lowering computational cost. Moreover, since computation across sub-bands are independent, the algorithm is naturally suited for parallel computing architecture.

Within each sub-band $\mathcal{B}^q$, we define a set of analysis bands, $\mathcal{A}_i^q \subset \mathcal{B}^q$, each characterized by a center frequency $f_{c_i}^q$ and bandwidth f_w^q ,

$$\mathcal{A}_i^q = \{f : |f - f_{c_i}^q| \leq f_w^q\}, \quad 0 \leq i \leq I^q - 1, \quad (17)$$

where I^q is the number of analysis bands in sub-band $\mathcal{B}^q$. These bands are identical, differing only by a frequency shift. The frequency resolution within each analysis band is $2f_w^q$, and the boundaries satisfy $f_{\min}^q = f_{c_{\min}} - f_w^q$, $f_{\max}^q = f_{c_{\max}} + f_w^q$. The full-signal band case is recovered when $Q = 1$, i.e., when the entire frequency range is treated as a single sub-band.

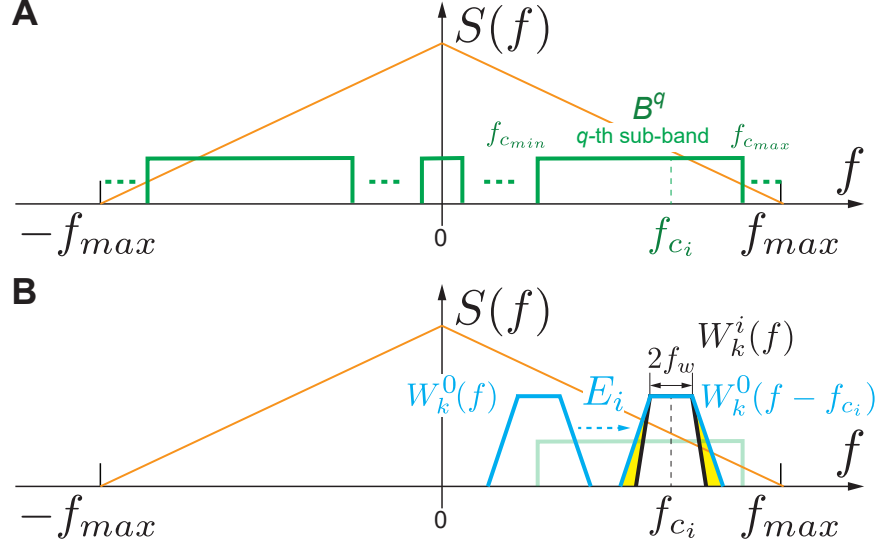


Figure 1: **Frequency-Domain Shift of Optimal Weighting Sequence.** Panel (A) illustrates the partitioning the signal band, defined as $B = \{f : |f| \leq f_{max}\}$, into multiple non-overlapping segments or sub-band. Each sub-band comprises a group of analysis bands, such as the q -th sub-band containing bands centered at frequencies f_{c_i} , ranging from $f_{c_{min}}$ to $f_{c_{max}}$. Panel (B) depicts the frequency-domain shift of the optimal weighting sequence, $W_k^0(f)$ (shown as a light blue trapezium), within the q -th band group. The shift moves the sequence from its original center at $f = f_{c_0}$ to a target analysis band centered at $f_{c_i}^q$. The bandwidth of each analysis band is $2f_w^q$ (superscript q omitted for clarity thereafter). The operator E_i denotes the frequency shift. The dark trapezium represents the frequency-domain transform of the optimal weighting sequence, $W_k^i(f)$, centered at f_{c_i} as per Bronez GPSS approach [39, 40]. Yellow shading highlights the sidelobe leakage difference—indicating suboptimality—between the optimal $W_k^i(f)$ and the shifted version $W_k^0(f - f_{c_i})$. The orange triangle represents the signal power spectrum, $S(f)$.

As discussed in Section 2, the primary computational bottleneck in BG method lies in the adaptive estimation of the tapers for each analysis band. To alleviate this computational burden, we propose computing the optimal tapers $\{\mathbf{w}_k^0\}_{k=1}^K$ only once at a nominal band $\mathcal{A}_0^q$ centered at $f_{c_0}^q$, and then shifting these tapers to all other analysis bands $\mathcal{A}_i^q$ within the sub-band B^q using the NuFFT (see Figure 1). This approach avoids repeated GEP solutions.

Since the analysis procedure is identical for each sub-band B^q , we omit the subscript q in the following derivations for clarity, unless otherwise specified.

3.2 Multitaper Nonuniform FFT (MTNuFFT) Estimator

In each sub-band B , we designate $\mathcal{A}_0$ centered at f_{c_0} as the nominal analysis band. The corresponding optimal tapers, denoted as $\{\mathbf{w}_k^0\}_{k=1}^K$, are determined by solving the GEP (14) at f_{c_0} ,

$$\mathbf{R}(\mathcal{A}_0)\mathbf{w}_k^0 = \lambda_k^0 \mathbf{R}(B)\mathbf{w}_k^0, \quad 1 \leq k \leq K, \quad (18)$$

where $\{\lambda_k^0, \mathbf{w}_k^0\}$ represents the k -th pair of eigenvalue and eigenvector for $\mathcal{A}_0$. The elements of the $N \times N$ positive definite Hermitian matrix $\mathbf{R}(\mathcal{A}_0)$ are given by (15)

$$\mathbf{R}(\mathcal{A}_0|n, m) = \int_{\mathcal{A}_0} e^{j2\pi f(t_n - t_m)} df$$

$$= \frac{\sin [2\pi f_w(t_n - t_m)]}{\pi(t_n - t_m)} \cdot e^{j2\pi f_{c_0}(t_n - t_m)}. \quad (19)$$

We define the frequency shifting operator E_i as

$$E_i = \begin{bmatrix} e^{j2\pi\Delta f_i t_1} & 0 & \cdots & 0 \\ 0 & e^{j2\pi\Delta f_i t_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{j2\pi\Delta f_i t_N} \end{bmatrix}, \quad (20)$$

where $\Delta f_i = f_{c_i} - f_{c_0}$. Note that if we choose the center frequency of the nominal band $\mathcal{A}_0$ at $f_{c_0} = 0$, then $\Delta f_i = f_{c_i}$. To void solving computationally costed GEP problem at $\mathcal{A}_i$, we may approximate the eigenvector $\mathbf{w}_k^i$ with $E_i \mathbf{w}_k^0$, which is shifted from $\mathbf{w}_k^0$, so that $J_k(\mathcal{A}_i) = \mathbf{w}_k^{i*} \mathbf{x}$, known as the eigencefficients [4], may be estimated as

$$\begin{aligned} \hat{J}_k(\mathcal{A}_i) &= (E_i \mathbf{w}_k^0)^* \mathbf{x} \\ &= \sum_{n=1}^N \left[w_k^{0*}(t_n) x(t_n) \right] e^{-j2\pi\Delta f_i t_n}. \end{aligned} \quad (21)$$

The power spectrum estimator of the integrated spectrum over analysis band $\mathcal{A}_i$ is then given by

$$\hat{P}(\mathcal{A}_i) = \frac{1}{K} \sum_{k=1}^K \left| \hat{J}_k(\mathcal{A}_i) \right|^2. \quad (22)$$

The eigencefficients (21) is typically implemented using Nonuniform FFT (NuFFT) [44, 42]. However, note that in sub-band $\mathcal{B}^q$, we use the relative frequency points Δf_{c_i} in calculating NuFFT. The desired frequency points are $f_{c_i} = f_{c_0} + \Delta f_{c_i}$. As for the special case, when the central frequency of $\mathcal{A}_0$ is $f_{c_0} = 0$, $\Delta f_{c_i} = f_{c_i}$. A relevant work to the NuFFT is the Nonuniform Discrete Fourier Transform (NDFT) [45] of a time series, which is defined as samples of its z -transform evaluated at distinct points located nonuniformly on the unit circle in the z -plane.

3.3 Bias Measure

We began by evaluating the performance of the estimator (22) in terms of bias. This assessment was carried out under the condition that the signal is white, meaning the true spectral density is flat, as previously employed in Bronez GPSS approach [40]. Specifically, we consider the case where $S(f) = 1$. The expectation of the estimator can be expressed as

$$\begin{aligned} \mathbf{E} \left[\hat{P}(\mathcal{A}_i) \right] &= \int_{\mathcal{B}} S(f) \frac{1}{K} \sum_{k=1}^K |W_k^i(f)|^2 df \\ &= \int_{\mathcal{B}} \frac{1}{K} \sum_{k=1}^K |W_k^0(f)|^2 df \\ &= \frac{1}{K} \sum_{k=1}^K \mathbf{w}_k^{0*} \mathbf{R}(\mathcal{B}) \mathbf{w}_k^0 \\ &= \mathbf{E} \left[\hat{P}(\mathcal{A}_0) \right], \end{aligned} \quad (23)$$

where we have replaced the DFT $\mathbf{W}_k^i(f)$ with $\mathbf{W}_k^0(f)$ according to the algorithm, and $\mathbf{w}_k^0$ is optimal at $\mathcal{A}_0$. Clearly, we have $\mathbf{E}[\hat{P}(\mathcal{A}_0)] = 2f_w$ by satisfying the normalization requirement (11). The bias of the estimator is then

$$\text{BIAS}\{\hat{P}(\mathcal{A}_i)\} = \mathbf{E}\left\{\hat{P}(\mathcal{A}_i)\right\} - \int_{f_{c_i}-f_w}^{f_{c_i}+f_w} S(f) df = 0. \quad (24)$$

3.4 Variance Bound

From (10), for Gaussian process, the bound factor may be written as

$$\begin{aligned} \mathbf{V}(\hat{\mathbf{w}}_1^i, \dots, \hat{\mathbf{w}}_K^i) &= \mathbf{V}(E_i \mathbf{w}_1^0, \dots, E_i \mathbf{w}_K^0) \\ &= \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K \left| \int_{\mathcal{B}} W_k^{0*}(f - f_{c_i}) \times W_l^0(f - f_{c_i}) df \right|^2. \end{aligned} \quad (25)$$

If the frequency center of the tapers are not near to the boundary of signal band $\mathcal{B}$, for instance, f_{c_i} and $\pm f_{\max}$ are separated by at least $2f_w$, (25) may be approximated as

$$\begin{aligned} \mathbf{V}(\hat{\mathbf{w}}_1^i, \dots, \hat{\mathbf{w}}_K^i) &\approx \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K \left| \int_{\mathcal{B}} W_k^{0*}(f) W_l^0(f) df \right|^2 \\ &= \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K \left| \mathbf{w}_k^{0*} \mathbf{R}(\mathcal{B}) \mathbf{w}_l^0 \right|^2 \\ &= \frac{(2f_w)^2}{K}, \end{aligned} \quad (26)$$

which is identical to the bound factor of the optimal approach.

3.5 Sidelobe Leakage and Suboptimality

As we shift the optimal eigenvectors (tapers) from the nominal analysis band $\mathcal{A}_0$ to $\mathcal{A}_i$, rather than using the optimal eigenvectors at the designated analysis band, it becomes crucial to understand the deviation from the optimal solution. As previously discussed, the bias measure and variance bound factor match the optimal ones, provided that the analysis band is not in proximity to $\pm f_{\max}$. We now consider the difference in bias bound factor between the optimal and our proposed solutions. We utilize this difference as a metric to indicate suboptimality of the fast algorithm (Fig. 1).

Using the identity $\mathbf{R}(\mathcal{A}_0) = E_i^* \mathbf{R}(\mathcal{A}_i) E_i$, where $\mathcal{A}_i = \mathcal{A}_0 + 2\pi(f_{c_i} - f_{c_0})$ (c.f., (3.42) in [39]), the bias bound factor can be expressed as

$$\begin{aligned} \mathbf{B}(\hat{\mathbf{w}}_1^i, \dots, \hat{\mathbf{w}}_K^i) &= \int_{\mathcal{B}} \frac{1}{K} \sum_{k=1}^K |W_k^q(f - f_{c_i})|^2 df - \frac{1}{K} \sum_{k=1}^K (E_i \mathbf{w}_k^q)^* \mathbf{R}(\mathcal{A}_i) (E_i \mathbf{w}_k^q) \\ &\approx \int_{\mathcal{B}} \frac{1}{K} \sum_{k=1}^K |W_k^q(f)|^2 df - \frac{1}{K} \sum_{k=1}^K \mathbf{w}_k^{0*} \mathbf{R}(\mathcal{A}_0) \mathbf{w}_k^0 \\ &= \frac{1}{K} \sum_{k=1}^K \mathbf{w}_k^{0*} [\mathbf{R}(\mathcal{B}) - \mathbf{R}(\mathcal{A}_0)] \mathbf{w}_k^0 \end{aligned}$$

$$= \frac{2f_w}{K} \sum_{k=1}^K (1 - \lambda_k^0). \quad (27)$$

We once again assume that $\mathcal{A}_i$ is not near the boundary of $\mathcal{B}$. The absolute value of bound factor difference can now be readily seen as

$$\begin{aligned} |\Delta \mathbf{B}_i| &\triangleq |\mathbf{B}(\hat{\mathbf{w}}_1^i, \dots, \hat{\mathbf{w}}_K^i) - \mathbf{B}(\mathbf{w}_1^i, \dots, \mathbf{w}_K^i)| \\ &= \frac{2f_w}{K} \left| \sum_{k=1}^K (\lambda_k^i - \lambda_k^0) \right| \\ &\leq \frac{2f_w}{K} \sum_{k=1}^K |\lambda_k^i - \lambda_k^0|. \end{aligned} \quad (28)$$

The difference (28) suggests that

$$\mathcal{E}_i = \frac{1}{K} \sum_{k=1}^K |\lambda_k^i - \lambda_k^0| \quad (29)$$

may serve as a measure of deviation from the optimal case. Clearly, $\mathcal{E}_i \in [0, 1]$ due to the eigenvalue condition $0 \leq \lambda_k^0, \lambda_k^i \leq 1$.

3.6 Thomson F -test for Nonuniform Signal

Statistical tests are often employed to ascertain the periodicity in signal. When a spectral peak is observed, it's crucial to determine if its magnitude significantly exceeds what could arise by chance. The Thomson F -test [8] serves as an effective tool for detecting spectral lines in colored noise (i.e., mixed spectrum), including biological signals [46, 6].

The F -statistic for the nonuniformly sampled time series may be formally computed from the eigenco-efficients (21). Assuming $2f_{c_i} > f_w$, the F -statistic can be derived as (c.f., pp. 496–500 in [4])

$$F_s = \frac{|\hat{C}_i|^2 (K-1) \sum_{k=1}^K [W_k^0(0)]^2}{\sum_{k=1}^K |J_k(\mathcal{A}_i) - \hat{C}_i W_k^0(0)|^2}, \quad (30)$$

where $W_k^0(0)$ is the NuFFT (21) of $\mathbf{w}_k^0$ at $\Delta f_i = 0$ (i.e., $f_{c_i} = f_{c_0}$), which is simply the summation of taper weights, $W_k^0(0) = \sum_{n=1}^N w_k^0(t_n)$. $\hat{C}_i$ is the estimated amplitude at f_{c_i} , calculated as

$$\hat{C}_i = \frac{\sum_{k=1}^K J_k(\mathcal{A}_i) W_k^0(0)}{\sum_{k=1}^K [W_k^0(0)]^2}. \quad (31)$$

The statistic in (30) follows an F -distribution, $F_s \sim F(2, 2K-2)$, with 2 and $2K-2$ degrees of freedom. The critical value F_α for a given level $\alpha = 1 - p$ can be found from the inverse F -distribution. As a general guideline, it is recommended [8, 4] to set the p -value at the Rayleigh frequency $1/N$, where N denotes the number of sample points.

Other studies proposed some similar statistical tests for spectral lines in nonuniformly sampled time series [37, 36, 21]. However, these methods generally do not normalize the tapers according to (11), which is essential for ensuring the energy conservation.

Table 3: M²NuFFT Algorithm of Spectral Estimation

1	Define sampling points $t_n, 1 \leq n \leq N$, time series samples $x(t_n), 1 \leq n \leq N$, signal band $\mathcal{B} = \{f : f \leq f_{\max}\}$, and the number of tapers K .
2	Define the <u>multiband</u> $\mathcal{B}^q = \{f : f_{\min}^q \leq f \leq f_{\max}^q\}$, nominal band $\mathcal{A}_0^q$ centered at $f_{c_0}^q$, and the half bandwidth f_w^q , for $q = 0, 1, \dots, Q - 1$.
3	FOR $q = 0, 1, \dots, Q - 1$ DO : IF $f_{c_0}^q = f_0 = 0$ DO : Derive ${}^q\hat{\mathbf{w}}_k^0 = \{{}^q w_k^0(t_1), \dots, {}^q w_k^0(t_N)\}$ by interpolation: (a) Compute DPSS on a uniform sampling grid, denoted as $v_n^{(k)}(N, f_w)$, where k is the order of the sequence. The grid interval is determined by the average inter-sample-interval $\bar{\Delta}_t = (t_N - t_1)/N$. (b) The taper weights $w_k^0(t_n)$ at intermediate points corresponding to the nonuniform times t_n are obtained by interpolation using a cubic spline. (c) Normalize the taper weights (eigenvector) such that $\hat{\mathbf{w}}_k^{0*} \mathbf{R}(\mathcal{B}) \hat{\mathbf{w}}_k^0 = 2f_w, 1 \leq k \leq K. \quad (11)$ ELSE DO : Find ${}^q\hat{\mathbf{w}}_k^0 = \{{}^q w_k^0(t_1), \dots, {}^q w_k^0(t_N)\}$ by solving the GEP: $\mathbf{R}(\mathcal{A}_0^q) \cdot {}^q\mathbf{w}_k^0(\mathcal{A}_0^q) = \lambda_k^{\mathcal{A}_0^q} \mathbf{R}(\mathcal{B}^q) \cdot {}^q\mathbf{w}_k(\mathcal{A}_0^q), \quad 1 \leq k \leq K, \quad (14)$ and normalize ${}^q\hat{\mathbf{w}}_k^0$. $\quad \quad \quad (11)$ END IF
4	Calculate the eigencoefficients by <u>NuFFT</u> : $J_k(\mathcal{A}_i^q) = \sum_{n=1}^N [{}^q w_k^{0*}(t_n) x(t_n)] e^{-j2\pi \Delta f_i t_n}, \quad 1 \leq k \leq K, 0 \leq i \leq I - 1. \quad (21)$
5	Compute the <u>multitaper</u> estimator of integrated spectrum: $\hat{P}(\mathcal{A}_i) = \frac{1}{K} \sum_{k=1}^K J_k(\mathcal{A}_i) ^2, 0 \leq i \leq I - 1. \quad (22)$
	END FOR

3.7 Computational Cost and M²NuFFT Algorithm

The computation of the optimal taper $\mathbf{w}_k^0$ necessitates the solution of the GEP at $\mathcal{A}_0$, as defined in (18). This step incurs a computational cost of $O(N^3)$, which is significant but performed only once per sub-band $\mathcal{B}^q$, unlike Bronez's original GPSS method, where the GEP must be solved for every analysis band. The transition of $\mathbf{w}_k^0$ to other analysis bands relies on the NuFFT, which demands $O(N \log N + N \log(1/\epsilon))$ arithmetic operations [41, 42], where ϵ is the precision of computation, without repeated GEP solutions.

Further computational savings are possible when the nominal analysis band $\mathcal{A}_0$ is centered at $f_{c_0} = 0$ Hz. In this special case, the optimal taper $\mathbf{w}_k^0$ may be approximated using the conventional DPSS $v_n^{(k)}(N, f_w)$, which are efficiently computable with fast algorithms [4, 47]. Given the sufficient regularity of the $v_n^{(k)}(N, f_w)$, it is advantageous to interpolate the uniformly sampled DPSS to nonuniform grid using a cubic spline [18] to circumvent the computation of GEP. Consequently, the overall computational cost of the fast M²NuFFT algorithm approximates that of NuFFT.

Instead of normalizing the power of $\hat{\mathbf{w}}_k^0$ to unity, we adhere to normalization requirements in equation (11). This normalization is of theoretical importance to ensure energy conservation when transforming between time and frequency domains. While alternative normalization methods—such as L^2 -norm normalization of the interpolated DPSS [18, 20]—may introduce a constant bias in the spectrum, the bias is often negligible in practice, as most applications focus on relative changes of power spectrum rather than absolute

values.

The complete M²NuFFT algorithm for spectral estimation in nonuniformly sampled time series is summarized in Table 3. Notably, the method proposed by Patil et al. [20] may be viewed as a special case of M²NuFFT, where $Q = 1$, the signal band $\mathcal{B}$ covering the entire frequency band, and $f_{c_0} = 0$. In this scenario, the computational complexity is effectively that of NuFFT. At the other extreme, when $Q = I$, where I is the total number of analysis bands, the M²NuFFT algorithm becomes equivalent to the original Bronez GPSS method with full frequency-dependent taper estimation.

Finally, the multiband structure described in Section 2.4 is inherently compatible with parallel computing architectures, enabling further reductions in computational time.

4 Taper Suboptimality Analysis

As previously discussed within the framework of Bronez GPSS [40], the suboptimality of the M²NuFFT algorithm primarily arises from two factors:

1. **Band mismatch:** The discrepancy between the optimal tapers $\mathbf{w}_k^0$, $k = 1, \dots, K$, defined at the nominal analysis band $\mathcal{A}_0$, and the optimal tapers $\mathbf{w}_k^i$ corresponding to the designated analysis band $\mathcal{A}_i$, for $i = 1, \dots, I - 1$.
2. **Interpolation error:** When an interpolated DPSS, denoted $\hat{\mathbf{w}}_k^0$, is used to approximate the optimal tapers at $\mathcal{A}_0$ (where $f_c = 0$) for computational efficiency, deviations between $\hat{\mathbf{w}}_k^0$ and the true optimal tapers $\mathbf{w}_k^0$ may contribute additional suboptimality.

Due to the complexity introduced by arbitrary sampling schemes, a closed-form analytical characterization of these discrepancies is beyond the scope of this work. Instead, we resort to numerical experiments to assess the suboptimality introduced by the M²NuFFT algorithm. Specifically, we computed the taper errors between M²NuFFT-generated tapers and the corresponding optimal tapers under four representative sampling scenarios: Uniformly Sampling, Jittering Sampling, Missing Data, and Arithmetic Sampling. The frequency domain is normalized to the interval 0–0.5 Hz, assuming a maximum signal frequency of $f_{\max} = 0.5$ Hz. All simulated signals have a fixed duration of $T = 50$ seconds.

We adopted the sampling schemes implemented in [39] to generate 50 timestamps. For uniform sampling, the samples were acquired at one-second interval, denoted as $t_n = n$, for $1 \leq n \leq 50$. To construct the jittering timestamps [48, 5], the sampling time was defined as $t_n = n + z_n$, where the jittered displacement process z_n was drawn from a Gaussian white noise (GWN) distribution with zero mean and a standard deviation of 0.1 seconds. The average sampling rate, or intensity [5], was set to 1 sample per second. For the missing-data sampling scheme, data were initially sampled at time points $t_n = 5n/6$, for $1 \leq n \leq 60$, followed by the random omission of 10 samples to simulate data loss. The fourth set of timestamps employed arithmetic sampling, defined as $t_n = 1 + a(n - 1) + b(n - 1)(n - 2)/2$, for $1 \leq n \leq 50$, where a and b are random variables governed by a control probability ρ [39].

For multitaper spectral estimation in M²NuFFT, we selected a bandwidth $f_w = 0.05$ Hz, yielding a frequency resolution of 0.1 Hz and a time-bandwidth (TW) of 2.5³. A total of $K = 4$ tapers were used for the analysis. For the Bronez GPSS methods (BGFixed and BGAdaptive), the signal band $\mathcal{B}$ was defined over the range 0–0.5 Hz. In BGFixed, the analysis bands $\mathcal{A}_i$ was centered at a frequency of interest f_{c_i} with a fixed bandwidth of 0.1 Hz, for $0 \leq i \leq I - 1$. In contrast, BGAdaptive dynamically adjusted both the number of tapers K and analysis bandwidth f_w . Initially, each analysis band was analyzed with four tapers and a bandwidth of 0.05 Hz. The number of tapers was then iteratively increased until the maximum side

³TW is conventionally defined as the product of the signal duration T and the bandwidth f_w , rather than the full bandwidth (frequency resolution) $2f_w$.

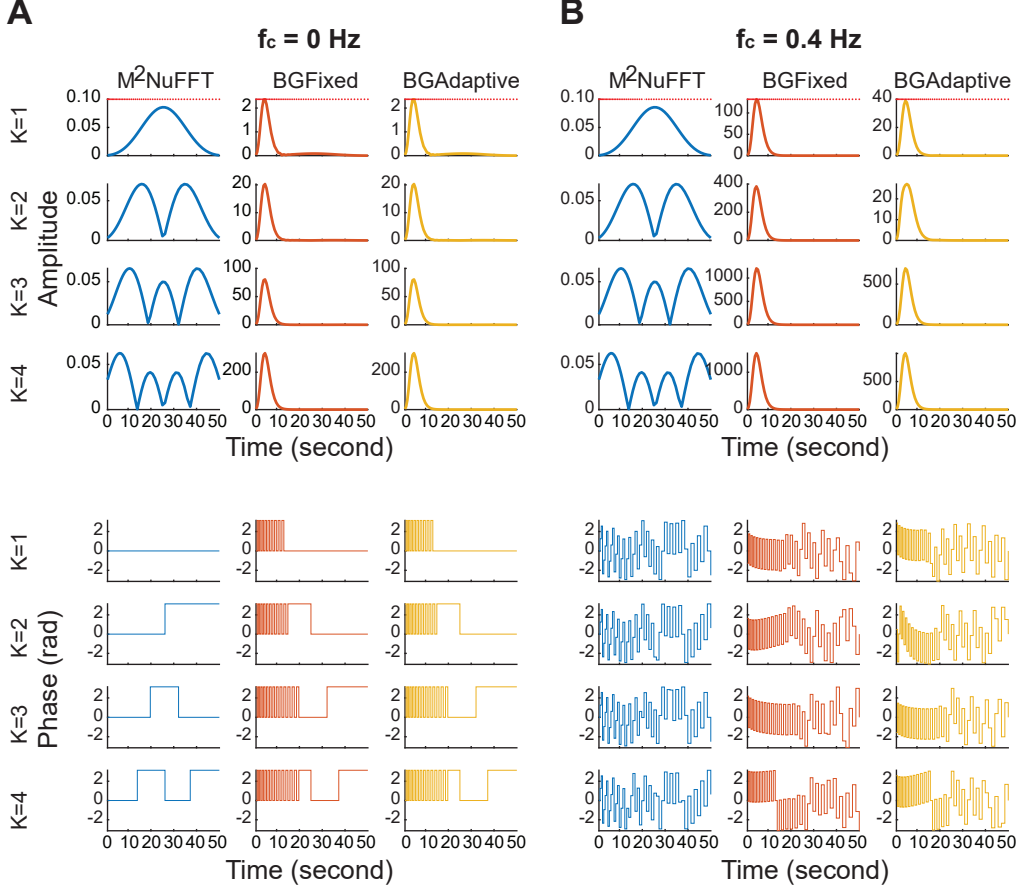


Figure 2: **Examples of Taper Amplitude and Phase for Arithmetic Sampling.** (A) Amplitude and phase of the first four tapers of the analysis band centered at $f_c = 0$ Hz, with a half-bandwidth of 0.05 Hz. The upper panel displays the amplitude, and the lower panel shows the phase. In each panel, rows correspond to a different taper, while columns represent the methods: M^2NuFFT (interpolated DPSS), BGFIXed, and BGAdaptive. Red dots at the tops of each column indicate an instance of arithmetic sampling. (B) As in (A), but for an analysis band centered at $f_c = 0.4$ Hz. Abbreviation: **M^2NuFFT** , multiband-multitaper nonuniform fast Fourier transform; **BGFIXed**, Bronez GPSS method with fixed TW [39]; **BGAdaptive**, adaptive Bronze GPSS method [39]; **DPSS**, discrete prolate spheroidal sequence [8].

lobe leakage⁴ of the tapers was less than -10 dB. If this criterion was not met after reaching the maximum of 8 tapers, the analysis bandwidth was incremented by 0.01 Hz, and the process repeated. Iteration continued until the leakage threshold was satisfied, or the bandwidth reached 0.5 Hz. The final number of tapers and analysis bandwidth were then used to estimate the power spectrum at the current frequency center. This adaptive process was repeated for all frequency centers.

Figure 2 illustrates example tapers in the time domain for the arithmetic sampling scheme, which exhibits the most pronounced discrepancies among the four sampling scenarios considered (see discussion below). The figure shows the amplitude and phase of the first four tapers at two analysis bands centered at $f_c = 0$ Hz and $f_c = 0.4$ Hz. At $f_c = 0$ Hz, M^2NuFFT tapers were approximated by interpolating the corresponding DPSS. At $f_c = 0.4$ Hz, the taper amplitude in M^2NuFFT remained identity to those at $f_c = 0$ Hz,

⁴For BGAdaptive, the *side lobe leakage* of taper k is calculated as $\gamma_k = 10 \log_{10} (1 - \lambda_k)$, where $1 \leq k \leq K$, which is given in TABLE I of Bronez, 1988 [40].

except for a phase modulation introduced by the NUFFT-based frequency shift. Note the higher sampling density at the beginning of the arithmetic sampling instance (red dots at the top of each column). In contrast, Bronez GPSS methods (BGFixed and BGAdaptive) adaptively adjusted the taper amplitude to local sampling variability, resulting in taper shapes that differ from those of M²NuFFT. Although the Bronez methods produced similar amplitude profile across the two analysis bands, the amplitudes scales differ besides phase modulation.

To quantify the deviation of M²NuFFT tapers from the optimal solution, we computed the root-mean-square error (RMSE) between the power spectra of M²NuFFT tapers and those of GPSS within $\pm f_w$ around each frequency centers f_{c_i} , for $i = 0, 1, \dots, I - 1$, of the analysis bands $\mathcal{A}_i$. We further assessed discrepancies using the eigenvalue differences defined in equation (29).

Figure 3 illustrates the average taper power spectra (computed as the DFT of the taper sequence; see equation (6)) for the four sampling schemes and three spectral estimation methods at the analysis bands centered at 0 Hz and 0.4 Hz. As expected, under uniform sampling (top row), the taper spectra are identical across methods. In contrast, arithmetic sampling (bottom row) exhibits substantial differences (c.f. Figure 2), reflecting the impact of nonuniform sampling on taper design.

We summarize the taper error analysis in Figure 4. Panel (A) shows taper power difference. For uniform sampling, no differences were observed since all methods produced identical tapers. For jittering and missing-data sampling, the errors remain relatively constant across the signal band, with missing-data sampling inducing significantly larger errors than jittering method. Interestingly, the arithmetic sampling errors increase approximately linearly with frequency, suggesting that the nearly symmetric tapers of M²NuFFT fail to capture local variations in sampling density at higher frequencies. Moreover, BGAdaptive generally exhibits larger errors than BGFixed, likely due to its lower sidelobe leakage achieved through adaptive procedure.

Panel (B) presents the eigenvalue difference analysis for BGFixed. The pattern mirrors that of Panel A—relatively constant differences for jittering and missing data sampling, and a linear increase for arithmetic sampling. This consistency supports the usage of equation (29) as a valid suboptimality metric. Note that the eigenvalue differences for BGAdaptive are not reported because the number of tapers varies cross analysis bands in the adaptive approach.

It is worth noting that the taper power difference analysis relied on interpolated DPSS to approximate the optimal tapers at the nominal band centered at 0 Hz, whereas the λ -difference analysis used the true optimal tapers at the same band. The strong similarity between these two sets of results suggests that the error introduced by DPSS interpolation is likely insignificant, at least for the sampling schemes considered in this study.

5 Performance Evaluation

Due to the complexity of the proposed M²NuFFT algorithm, in this section we only consider the performance on the full signal band, i.e., $B = \{f : |f| \leq f_{\max}\}$, and the nominal band $\mathcal{A}_0$ centered at zero frequency, which is usually the case in practice. We evaluated the performance in three key aspects: accuracy, speed, and real-world applicability. Initially, we computed the Mean-Square Error (MSE) between the estimated and the actual power spectra of Gaussian white noise (GWN) under various sampling schemes. The results indicated that the error range of M²NuFFT was compatible with that of the optimal method, BGAdaptive. Subsequently, we contrasted the speed of M²NuFFT with three alternative methods. Our findings revealed that the speed of our algorithm is 2–3 orders of magnitude higher than that of the optimal method. Lastly, we applied our method to estimate the power spectrum of a real-world signal, specifically a nonuniformly sampled impedance measurement. We then compare the outcomes of Thomson’s F-test on the periodicity of both the original and resampled signals. This comparison allows us to evaluate the

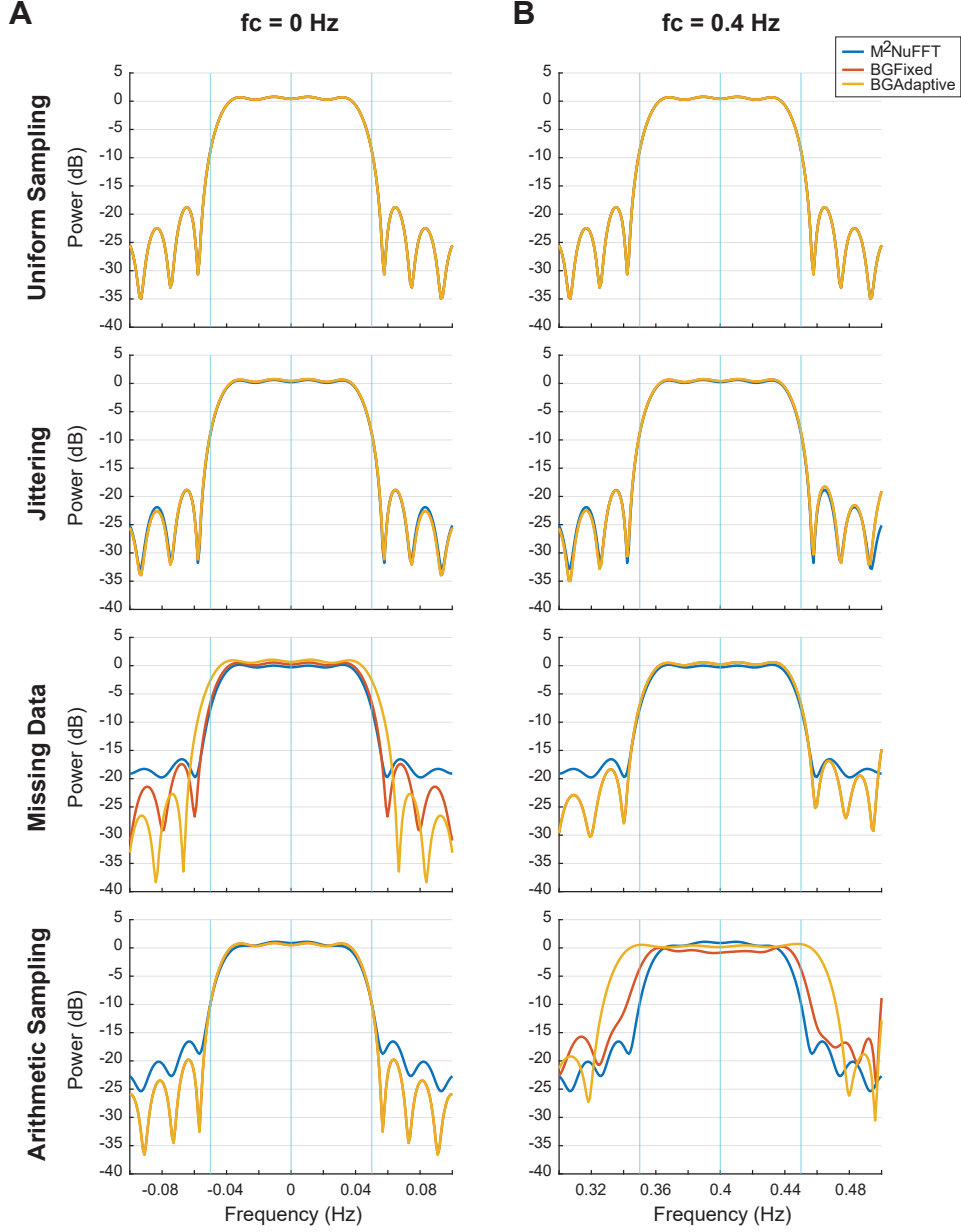


Figure 3: **Examples of Taper Power Spectra Across Sampling Methods.** (A) Average power spectra for four sampling schemes—Uniform sampling, Jittering, Missing data and Arithmetic sampling—estimated using three power spectrum estimation methods: M^2NuFFT , $BGFxed$ and $BGAdaptive$ at the analysis band centered at $f_c = 0$ Hz with a half-bandwidth of 0.05 Hz. The three cyan vertical lines indicate the center and the bound of the analysis band. (B) The same as in (A) except $f_c = 0.4$ Hz. Abbreviation: M^2NuFFT , multiband-multitaper nonuniform fast Fourier transform; $BGFxed$, Bronez GPSS method with fixed TW [39]; $BGAdaptive$, adaptive Bronze GPSS method [39]; $DPSS$, discrete prolate spheroidal sequence [8].

effectiveness of our method in practical applications.

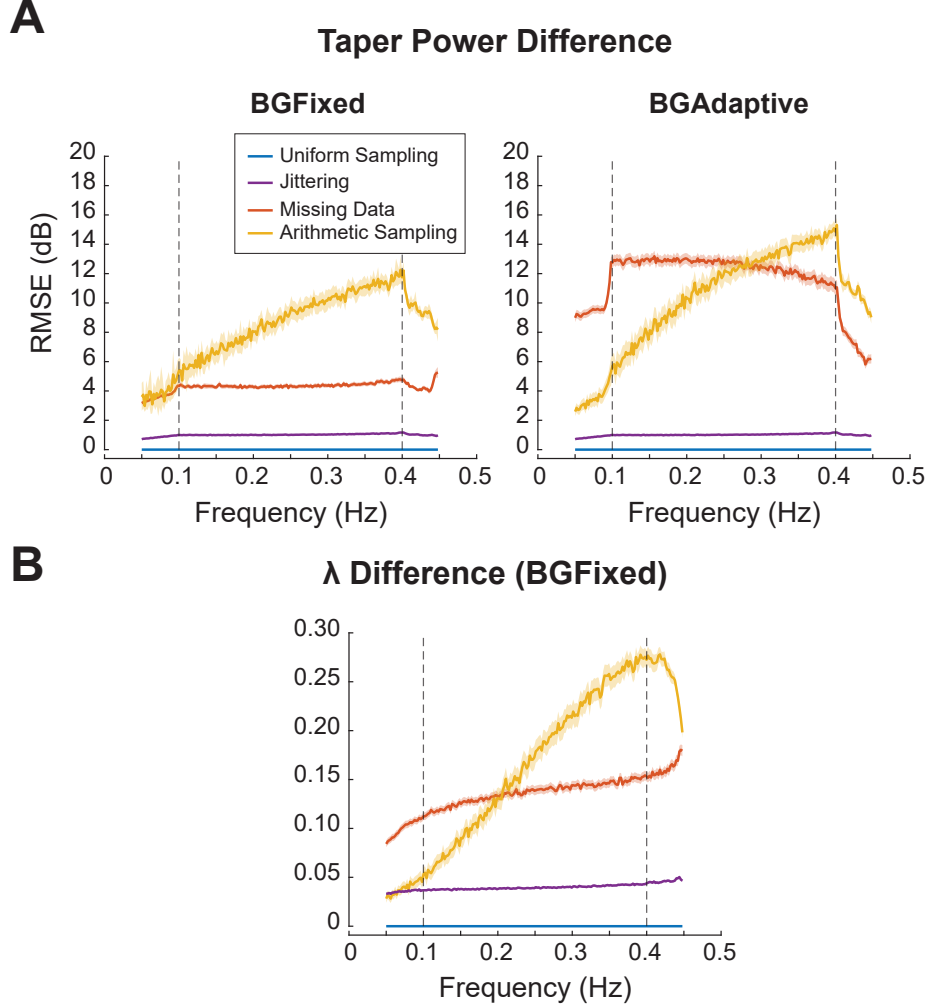


Figure 4: **M²NuFFT Tapper Error Analysis.** (A) *Taper power difference.* RMSE of the power difference between tapers estimated by M²NuFFT and Bronez GPSS method—BGFIXED (left panel) and BGADAPTIVE (right panel)—across four sampling schemes: Uniform sampling, Jittering, Missing data and Arithmetic sampling. Dashed vertical lines indicate the centers of the analysis bands at 0.1 Hz and 0.4 Hz, which are located $2f_w$ away from the edges of the signal band. Estimated errors outside the dashed lines may be unreliable due to the boundary effect. (B) *λ difference [see Eq. (29)].* Estimated average difference between the eigenvalues of the optimal GPSS tapers at the analysis band centered at 0 Hz and those centered at f_{c_i} , using BGFIXED method. Dashed vertical lines are as in (A). Shaded error band: ± 1 standard error of the mean (SEM). Abbreviation: **RMSE**, root-mean-square error; **M²NuFFT**, multiband-multitaper nonuniform fast Fourier transform; **BGFIXED**, Bronez GPSS method with fixed TW [39]; **BGADAPTIVE**, adaptive Bronze GPSS method [39]; **GPSS**, generalized prolate spheroidal sequence [8].

5.1 Error Analysis

We assessed the accuracy of the proposed fast algorithm by comparing the estimated spectrum with the true spectrum of unit variance GWN ($\sigma^2 = 1$), which was conducted using four estimation methods (MTLS, BGFIXED, BGADAPTIVE, and M²NuFFT) and the four sampling schemes described above (Section 4) to sample 50 points from the GWN.

The multitaper methods adopted in M²NuFFT, BGFIXED and BGADAPTIVE were described in Section 4.

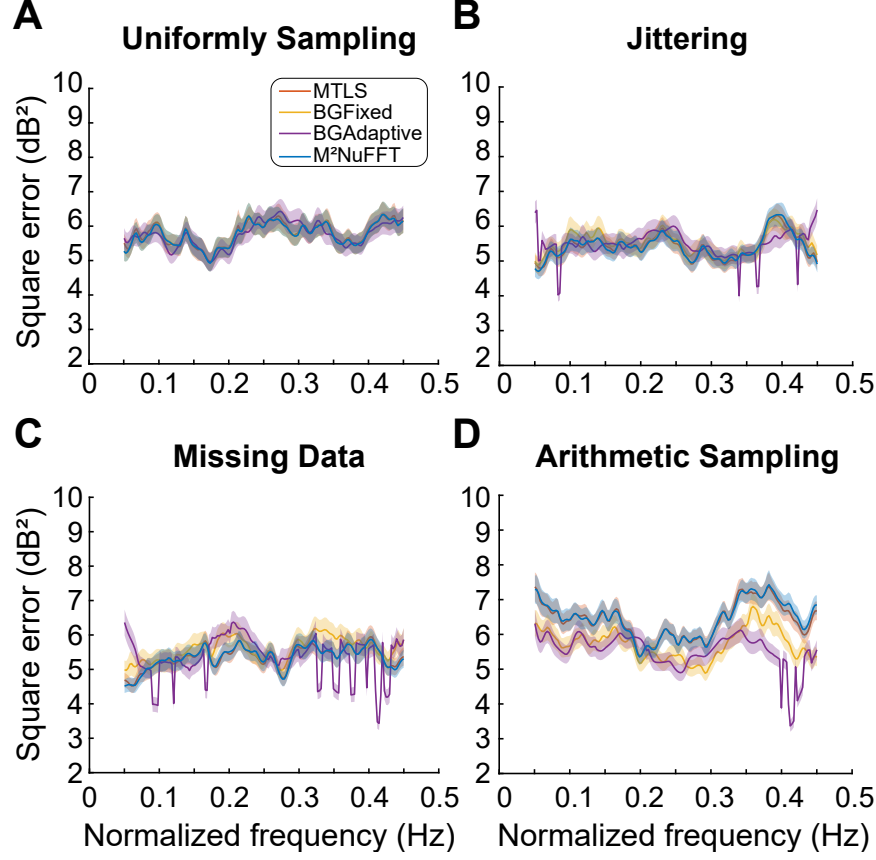


Figure 5: Error Analysis of Spectrum Estimation Methods. Mean-square error comparison between the true spectrum (Gaussian white noise, GWN) and the estimated spectra using four estimation approaches, i.e., MTLS, BGFxed, BGAdaptive, and M²NuFFT, and the signals were sampled using four sampling schemes: (A) Uniformly Sampled, (B) Jittering, (C) Missing Data, and (D) Arithmetic Sampling. The frequency range was normalized to 0–0.5 Hz and half-band width (f_w) was set at 0.05 Hz. Error measures at 0–0.5 and 0.45–0.5 Hz were omitted due to unreliable estimation. Error band: ± 1 SEM. Abbreviation: **MTLS**, multi-taper Lomb-Scargle periodogram [18]; **BGFxed**, Bronez GPSS method with fixed TW [39]; **BGAdaptive**, adaptive Bronez GPSS method [39]; **M²NuFFT**, multiband-multitaper nonuniform fast Fourier transform (* proposed in this article).

The parameters of the tapers for MTL were identical to those of M²NuFFT.

We repeated the process to evaluate the power spectrum of the GWN $M = 1000$ times for each estimation method and each sampling scheme. Subsequently, we computed the MSE in decibels (dB) between the estimated spectrum $\hat{S}(f_{c_i})$ and true spectrum $S(f_{c_i})$ at each frequency center f_{c_i} , which was calculated as $\text{MSE}(f_{c_i}) = \frac{1}{M} \sum_{m=1}^M [10 \log_{10} \hat{S}(f_{c_i})]^2$, given that $S(f_{c_i}) = 1$.

Figure 5 presents the error analysis results, organized into four panels that corresponds to the four sampling schemes. Each panel displays the mean and standard error of the mean (SEM) of squared error at each frequency center. For uniformly sampled signal, the error range was essentially identical across all four estimation methods. The M²NuFFT method demonstrated a compatible error range to the Bronez GPSS methods (BGFxed, BGAdaptive) when applied to jittering and missing-data sampling. However, the BGAdaptive method exhibited a better performance at isolated frequency centers. In the case of the arithmetic sampling scheme, the Bronez GPSS methods marginally yet significantly outperformed both MTLS

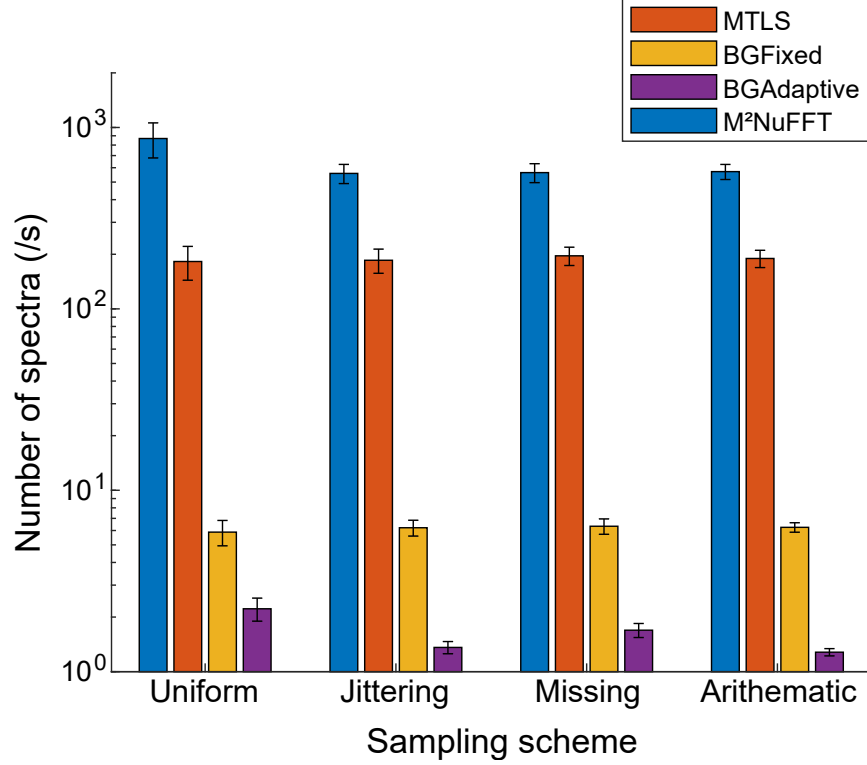


Figure 6: **Speed Analysis of Spectrum Estimation Methods.** This figure presents the number of spectra calculated per second for four sampling schemes using four spectrum estimation methods. The abbreviations used are consistent with those in Figure 5. Error bar: ± 1 STD

and M²NuFFT across the entire signal band $\mathcal{B}$. Overall, the proposed fast algorithm M²NuFFT demonstrated competitive accuracy in spectrum estimation in three of the four sampling schemes investigated.

5.2 Speed Analysis

The time efficiency of the proposed method was assessed by comparing the number of spectra computed per second across four different sampling schemes, using the four spectrum estimation methods. The performance evaluation was conducted on a Windows 10 HP workstation equipped with an Intel Core i5–10500 CPU operating at 3.10 GHz and 64 GB memory. We estimated the time cost for 1,000 spectrum estimation and obtained the mean and standard deviation (STD). The results, as presented in Figure 6, indicate that the speed of M²NuFFT significantly surpasses that of MTLS and 2–3 orders of magnitude faster than the Bronez GPSS approaches for all four sampling schemes investigated.

5.3 Application to Impedance Signal

To further illustrate the M²NuFFT method, we applied it to the spectral analysis of a bio-impedance signal recorded intracranially from human brain using a chronically implanted sensing and stimulation device. The specifics of the brain impedance acquisition and analysis have been detailed in our prior work [27, 28]. The impedance signal was measured using the investigational Medtronic Summit RC+S™ device with the electrodes targeting the limbic system of a patient with epilepsy. Given that the same electrodes were also utilized for delivering electric stimulation as part of neuromodulation therapy, the impedance measurements were nonuniformly sampled at an approximate rate of one sample every 15 minutes, equivalent to about

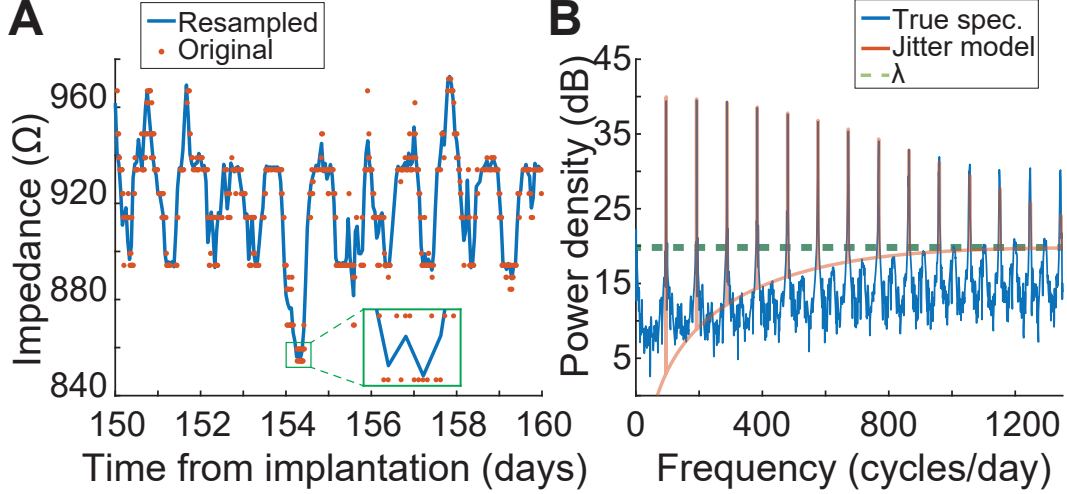


Figure 7: **Impedance Signal Sampling and Jitter Model Fitting.** (A) An example of impedance signal nonuniformly sampled at approximately four samples per hour using a Medtronic Summit RC+STM neuro-modulation device. Red dots represent original samples obtained 150–160 days post-implantation. The blue curve shows the resampled signal at a uniform rate of one sample per hour. The green rectangular inlet is a zoomed signal interval, indicating highly packed impedance samples (red dots). (B) Power spectrum of the sampling points from (A) and its fitted model. The blue curve represents the sampling process spectrum (calculated with CHRONUX function `mtspectrump`, $TW = 3.5$ and $K = 6$), while the red curve depicts the fitted model, a uniformly sampled point process with jitter (see text for more details). The green dashed line marks the average sampling rate ($\lambda = 96$ /day or $10 \log_{10}(96) = 19.82$ dB). Abbreviation: **spec**, spectrum.

96 samples/day. Figure 7A shows a data segment of between 150 and 160 days post device implantation (number of sample, $N = 688$), which was used in the analysis. The red dots represent the original impedance samples, while the blue curve signifies the resampled signal at a uniform rate of one sample per hour ($N = 240$, calculated with MATLAB function `resample` using linear interpolation). In Panel B, the blue curve illustrates the power spectrum of the *sampling instances* of the original impedance signal. The decaying envelope of the sharp lines at the fundamental frequency of 96 cycles/day and its harmonics are indicative of irregular sampling [48, 49, 50]. We fitted the spectrum with a jittering model [5], assuming a normal distribution of the jittering displacement z_n with zero mean and STD σ . The red curve in Figure 7B represents the fitted model with mean rate $\lambda = 96$ samples/day and STD $\sigma = 20$ seconds, and the green dashed line indicates the mean rate at high-frequency limit. This model offers a good understanding of sampling properties of the impedance measurement sequence.

Subsequently, we computed the power spectra of the original signal and the resampled signal using M²NuFFT and (Chronux) `mtspectrump`, respectively, under the assumption of a maximum frequency of 12 cycles/day. Identical to the calculation of point process power spectrum, the TW was set at 3.5, yielding a frequency resolution of 0.35 cycles/day, and K at 6. As displayed in Figure 8A, the spectral power of the nonuniformly sampled signal (represented by the red curve) is noticeably elevated above approximately 2 cycles/day in comparison to the spectral power of the resampled signal (blue curve). This observation aligns with previous studies [18, 35] (see Section 1: Introduction). While the spectral powers are nearly identical around the circadian cycle (1 cycle/day, indicated by arrow a) and multiday cycles (< 1 cycle/day), distinct energy peaks are presented in the frequency ranges of 2–4 Hz (arrow b) and 4–6 Hz (arrow c), which are absent in the power spectrum of the resampled data. However, the elevated power at high-frequency range of 10–12 Hz (arrow d) could potentially be attributed to leakage (refer to Section 6 Discussion).

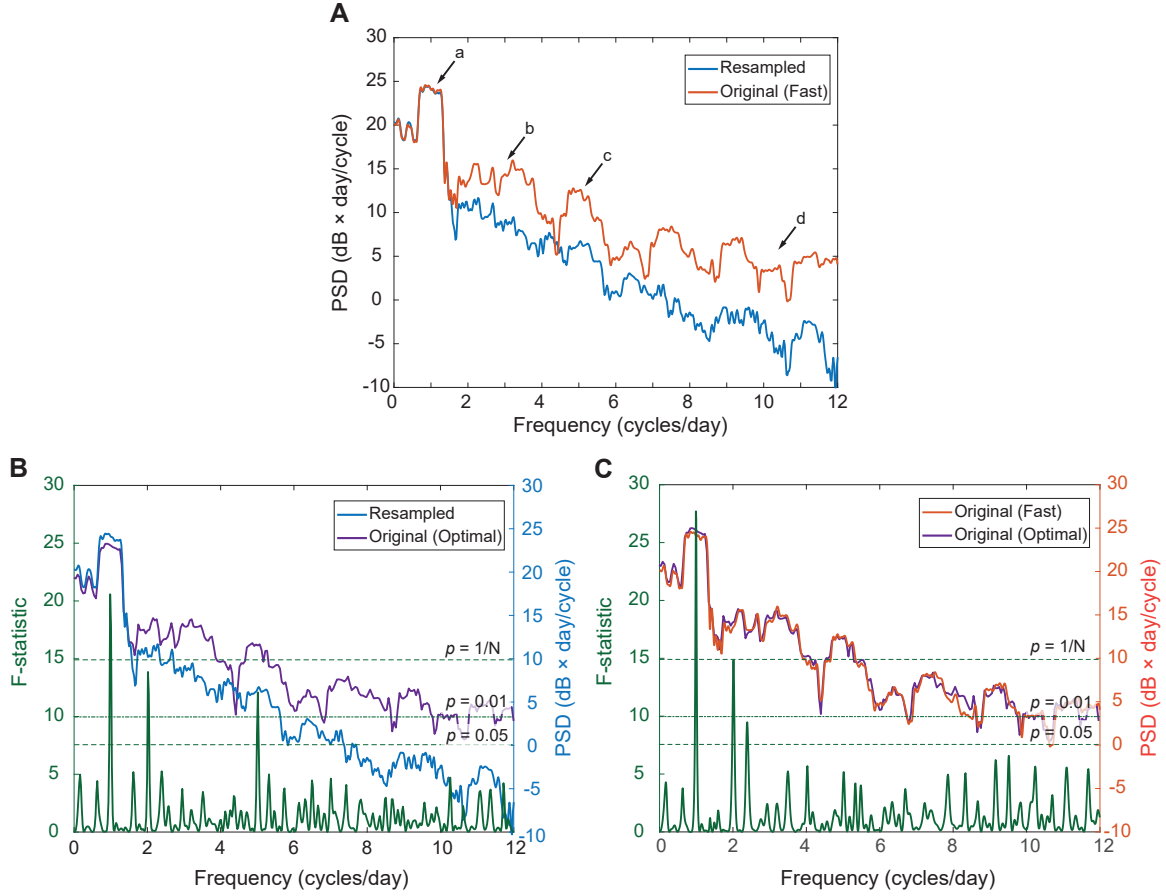


Figure 8: Spectrum and F-test Comparison. (A) Spectrum comparison. The red curve (Original (Fast)) represents the estimated power spectrum of the nonuniformly sampled impedance data (referenced as red dots in Fig. 7A) using MTNUFFT. The blue curve (Resampled) depicts the power spectrum of the resampled signal (blue curve in Fig. 7A), estimated with the CHRONUX [51] function `mtspectrumc`. The arrows point to approximate frequency bands of circadian cycles (a, 1 cycle/day), 2–4 (b), 4–6 (c), and 10–12 cycles/day (d). (B) Power spectrum of the resampled signal (blue, same as in Panel A), power spectrum of the original signal estimated with the optimal method `BGFixed` (purple, original (Optimal)), and Thomson periodicity F-test [8] (green) of the resampled signal. (C) Power spectrum estimated with the fast method (red, same as in Panel A, original (Fast)), power spectrum estimated with the optimal method (purple, same as in Panel B, Original (Optimal)), and F-test (green) of the original, nonuniformly sampled signal. In Panels B and C, three horizontal dashed lines represent three levels of p -values, i.e., from bottom to top, 0.05, 0.01 and $1/N$, respectively, where N is the number of samples. For the resampled signal (Panel B), $N = 240$, resulting in $p = 0.0042$. For the original signal (Panel C), $N = 688$, yielding $p = 0.0015$.

Moreover, we investigated the periodicity of impedance data using the Thomson F -tests [8, 4]. Specifically, we analyzed both the original (Figure 8B) and resampled data (Figure 8C), where the optimal estimation of the spectrum using `BGFixed` is superimposed for comparison. To assess significance, we calculated critical values of the F -statistic corresponding to three significant levels: p -values at 0.05, 0.01 and the Rayleigh level ($1/N$, where N is the number of samples). These critical values were derived from the

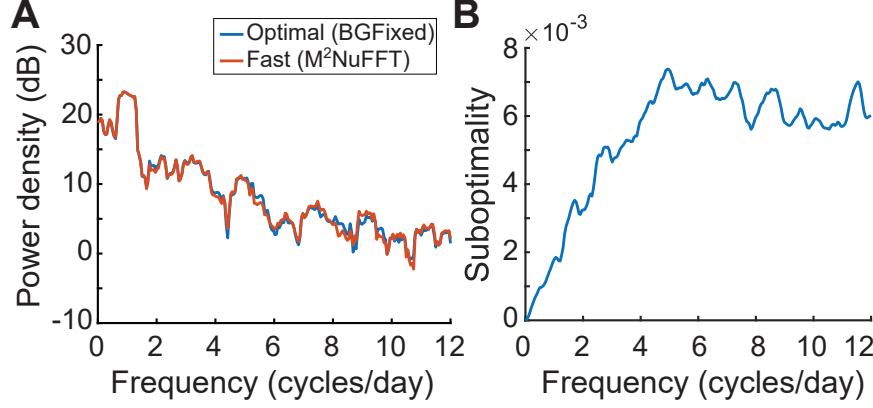


Figure 9: **Spectrum comparison and suboptimality.** (A) The optimal spectrum estimated with BGFixed method (blue) is compared with the spectrum estimated with the fast algorithm (red, M²NuFFT), where the eigenvectors were $\mathbf{w}_k^0$, $1 \leq k \leq K$ (18), optimal at $\mathcal{A}_0$. (B) Suboptimality (29) of the spectral estimation M²NuFFT shown in Panel A.

F -distribution with 2 and $2K - 2 = 10$ degrees of freedom. It is worth noting that the Rayleigh level ($p = 1/N$), as recommended in Thomson et al. [8], bears similarity to the Bonferroni correction for multiple comparison [52, 53]. The analysis reveals some intriguing findings. The F -statistic for the resampled signal (Panel B) indicates a strong periodicity in the circadian cycle (above the Rayleigh level) and suggests two possible cycles around 2 and 5 cycles/day (above $p = 0.01$ level). In contrast, the F -statistic for the original signal confirms the robust periodicity of the circadian cycle and its harmonic at 2 cycles/day (above the Rayleigh level). Notably, it also depicts the absence of the periodicity around 5 cycles/day, raising the possibility that this suggested periodicity in the resampled signal may result from linear interpolation.

These analyses of the ultradian cycles (occurring more frequently than once per day) within impedance signals hold significant biological interest. Long-standing hypotheses propose that an ultradian basic rest-activity cycle (BARC) plays a crucial role in sleep cycles, wakefulness patterns, and the central nervous system functioning [54].

Finally, we evaluated the suboptimality of the power spectrum estimation of the impedance signal by calculating $\mathcal{E}_i$ (29). Given that the approximate eigenvectors $\hat{\mathbf{w}}_k^0$ lack corresponding eigenvalues, we utilized the true eigenvectors $\mathbf{w}_k^0$ at $\mathcal{A}_0$ for the fast algorithm M²NuFFT (see Table 3), denoted as M²NuFFT0. The power spectra, estimated by both the optimal method (BGFixed) and the fast algorithm (M²NuFFT0), are depicted in Figure 9A. It's important to note the identity of the spectra at $f = 0$, which is confirmed by the suboptimality measure $\mathcal{E} = 0$ at $f = 0$, as shown in Figure 9B. As previously discussed, suboptimality $\mathcal{E}$ is between 0 and 1, where zero signifies optimality. The larger this measure, the greater the deviation of the estimation is from the optimal scenario. The suboptimality increases as the center of the analysis band shifts away from $f = 0$, but appears to plateau after about $f = 5$ cycles/day. Overall, the suboptimality is less than 8×10^{-3} , suggesting that the proposed method effectively estimated the power spectrum of the impedance signal.

6 Discussion

We have developed M²NuFFT, a method for a rapid and scalable power spectrum estimation of nonuniformly sampled time series. This method alleviates the computational burden of the Bronez GPSS in three key aspects: (1) The multiband framework enables partitioning of the signal band into sub-bands, which is well suited for parallel computing architecture, allowing substantial computational time reduction. (2)

The algorithm requires solving the computationally intensive GEP only once per sub-band. The resulting optimal tapers $\mathbf{w}_k^0$ for $\mathcal{A}_0$ are then shifted to other analysis bands $\mathcal{A}_i$, for $1 \leq i \leq I - 1$, using the NuFFT to circumvent the remaining GEP problems. (3) In the special case when $\mathcal{A}_0$ is centered at zero frequency, the optimal taper $\mathbf{w}_k^0$ can be efficiently approximated by interpolating the DPSS tapers, $v_n^{(k)}(N, f_w)$, to the nonuniform grid using cubic splines [18]. This eliminates the need for GEP entirely. As a result, the overall computational cost aligns with the fast NuFFT, $O(N \log N + N \log(1/\epsilon))$, which is significantly lower than the $O(N^4)$ complexity of the optimal method. Simulations (Figure 6) show that the computing speed of M²NuFFT is significantly higher than Multitaper Lomb-Scargle periodogram (MTLS), and 2–3 orders of magnitude faster than the BG methods (both BGFixed and BGAdaptive) in the zero-centered case.

Bronez [39, 40] also proposed a computationally efficient approximation known as the Constrained-Basis Weighting Sequences method. The basic idea is to reduce the size of the matrix in the GEP (14) from N to M , ideally $M \ll N$. This is achieved by selecting M “basis vectors” and approximating the weight sequence as $\hat{\mathbf{w}}_k(\mathcal{A}) = F \cdot \mathbf{c}_k(\mathcal{A})$, where F is an $N \times M$ matrix with columns as the predefined basic vectors, and $\mathbf{c}_k(\mathcal{A})$ are $M \times 1$ vectors determined by solving another GEP for analysis band $\mathcal{A}$. However, the matrix size of this problem is only M . The vector $\mathbf{c}_k(\mathcal{A})$ still needs to be calculated for each analysis band. The overall computing load is in the order of $O(M^2 N^2)$, which can be significantly lower than $O(N^4)$, but still notably higher than that of M²NuFFT. The choice of the basis vectors is critical to the method’s performance and requires careful consideration.

Since M²NuFFT is not an optimal solution, it is crucial to evaluate the deviation from the optimal solution of its estimates. Our theoretical work (Section 3) and numerical experiments (Section 4) show that the bias of estimation and variance bound are compatible with the optimal Bronez GPSS. But the bias bound is generally degraded [55]. The suboptimality of M²NuFFT for each analysis band $\mathcal{A}_i$ may be quantified by the difference between λ_k^0 and λ_k^i (29), which decreases at the expense of increasing analysis band (decrease of frequency resolution). The simulation results (Figure 5) show that for the four sampling schemes under investigation, the error range is compatible with the optimal method for uniformly sampling, jittering and missing-data sampling. Only the error range for arithmetic sampling is consistently higher than the optimal methods. These results indicate the effectiveness of the proposed method for most practical scenarios. Although our analysis attributes the suboptimality less to the interpolation of DPSS than to the frequency shifting of tapers via NuFFT, the likely loss of orthogonality of the interpolated DPSS remains an interesting topic for future research [56].

It is worth noting that the variance and bias bounds may not be accurately estimated when the analysis bands are near the boundary of signal band, $f_{\max}$. In deed, a previous study [57] showed that the variance of spectrum estimates could be poorly estimated if the frequency was close to frequency limits.

As noted in Section 3, the algorithm proposed by Patil et al. [20, 21], referred to as mtNUFFT, corresponds to a special case under our more generalized M²NuFFT framework. Their work provided compelling empirical evidence for the practical effectiveness of their algorithm, particularly for astrophysical data analysis. However, our contribution extends beyond empirical validation by offering a theoretical framework that quantifies the estimator’s performance in terms of bias, variance and suboptimality, which were not sufficiently addressed in prior work. This theoretical development allows us to recognize that mtNUFFT and Bronez GPSS method represent two extremes under the same M²NuFFT framework. At one end, mtNUFFT assumes a single signal band centered at zero frequency with interpolated DPSS tapers; at the other, Bronez’s method computes frequency-dependent tapers for each analysis band. Our framework bridges these approaches, offering a continuum of trade-offs between computational efficiency and statistical optimality.

Moreover, our normalization approach, based on equation (11), ensures signal energy conservation (c.f. Figure 3). This insight is not readily apparent through numerical evaluation alone. In contrast, the L^2 -norm normalization commonly used for uniformly sampled DPSS may introduce a constant bias in the power

spectrum estimate. While this bias may be negligible in practice, it underscores the importance of theoretically grounded normalization. Patil et al. also emphasize the importance of quasi-regular time sampling for the effectiveness of mtNUFFT. Our analysis provides a theoretical rationale for this observation. As seen in Section 4 and 5, jittered and missing-data sampling tend to produce a relatively constant and smaller deviation from the optimal solution (Figures 4 and 5). These sampling patterns preserve the structure of the signal sufficiently well to maintain taper performance, making them favorable for suboptimal methods like M^2 NuFFT.

The process of resampling nonuniformly sampled values to a uniform grid, for example, using linear interpolation, is often employed for spectral analysis due to the powerful tools available for estimators under uniform sampling. However, evidence suggests that this procedure could induce considerable artifacts in the power spectrum [35]. One noticeable effect of linear interpolation is a tendency for the estimated spectrum at high frequency to be lower, and at low frequency to be higher [18]. This distortion can be substantial [58]. Our analysis of the brain tissue impedance data also indicates a suppression of spectral power at higher frequencies due to interpolation. As shown in Figure 8A, the spectrum estimated from original samples using M^2 NuFFT (red curve) and the spectrum estimated from the resampled signal (blue curve) are nearly identical at lower frequency ranges (< 1.5 cycles/day), but notably different at higher frequency ranges (1.5–12 cycles/day). This observation is consistent with the previous findings and underscores the need to develop spectral estimators that directly utilize nonuniformly sampled data.

The M^2 NuFFT algorithm may be considered as a general framework for quickly estimating the spectrum of nonuniformly sampled signals using various types of tapers. Besides DPSS sequences, other tapers, such as minimum bias tapers and sinusoidal tapers [59, 60], have been previously suggested for different problems. Generally, these methods have not been extended to the case of nonuniformly sampled signals. By replacing the weight sequence w_k^0 in (21) with the desired tapers, which are properly evaluated on the nonuniform sampling grid, and then applying the NuFFT and averaging (22), the M^2 NuFFT algorithm may be extended to these tapers for spectrum estimation in nonuniformly sampled time series. Future work will focus on extending M^2 NuFFT to different tapers in various data analysis scenarios and evaluating the statistical properties of the estimation.

A limitation of this study is that the theoretical analysis of M^2 NuFFT optimality relies on minimizing the estimation bounds of Gaussian white noise spectrum. While this optimality criterion has proven effective for evaluating the performance of the fast algorithm in neurophysiological signal spectral estimation—our main targeted field of application—it may not be universally suitable. Specifically, an optimality criterion based on colored noise could be more appropriate for other applications. Future work will focus on extending the theoretical analysis to these scenarios.

The current M^2 NuFFT algorithm assumes that the partitioning of sub-signal bands, $\mathcal{B}^q$, and the bandwidth, f_w (determined by the number of tapers K), are pre-defined (see Table 3). However, the proposed framework suggests a potential pipeline for adaptive and iterative refinement of these parameters. As a first step, a cluster analysis of the sampling times could be performed to identify regions of relatively dense sampling, separated by large temporal gaps. Within each cluster, an initial spectral estimate can be obtained using a fast implementation of M^2 NuFFT (e.g., $Q = 1$ and $f_0 = 0$) based on an initial guess of sub-band structure and bandwidth. Subsequent refinement can then be guided by the characteristics of the preliminary spectrum: for regions where the spectrum is smooth and slowly varying, fewer and wider sub-bands (i.e., decrease Q and increase f_w) may suffice; conversely, regions exhibiting rapid spectral changes may benefit from finer segmentation (i.e., increase Q and decrease f_w). This adaptive procedure can be repeated iteratively to improve spectral estimates within each cluster. For critical frequency regions—such as those associated with critical physiological or physical phenomena—the Bronez GPSS method can be selectively applied to obtain optimal spectral estimates for the corresponding sub-bands. Finally, the spectral estimates from all clusters can be aggregated to produce a comprehensive and refined power spectrum. A fully automated implementation of this iterative refinement strategy represents an interesting direction for future

work.

7 Summary

This paper introduces M²NuFFT, a fast suboptimal method for power spectrum estimation in nonuniformly sampled time series. The proposed multiband-multitaper framework leverages parallel computing architecture and avoids the computational bottlenecks of Bronez GPSS method. In each sub-signal band, the estimator comprises a set of nonuniformly sampled tapers, optimized for a nominal analysis band. The estimated power within the band is determined by averaging the power correlated to these tapers. The NuFFT is utilized to swiftly shift the tapers to other analysis bands of interest, thereby removing the need to solve the GEP repeatedly. In the special case where the nominal band is centered at zero frequency, the GEP computation can be circumvented entirely by approximating the GPSS tapers with the interpolated DPSS tapers. The overall computational complexity of M²NuFFT in this case aligns with that of NuFFT, $O(N \log N + N \log(1/\epsilon))$.

The statistical properties of the estimator are assessed using the Bronez GPSS theory. The results reveal that the bias of the estimates and variance bound of M²NuFFT are comparable to those of the optimal estimator. However, the limitation of M²NuFFT lies in the degradation of the bias bounds. The difference in bias bounds between M²NuFFT and the optimal estimator may serve as a measure of suboptimality. Simulation results indicate that M²NuFFT operates 2–3 orders faster than the optimal method. Moreover, the error range of M²NuFFT aligns with that of the optimal estimator in three out of four sampling schemes under investigation, suggesting effectiveness in practical applications. The M²NuFFT together with the proposed extension of Thomson F -test, is suitable for rapid spectrum estimation and periodicity testing in large nonuniformly sampled datasets for exploratory analysis.

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A GPSS Matrix of Sub-Signal Band is Positive Definite Hermitian

According to the definition shown in (9), the Generalized Prolate Spheroidal Sequence (GPSS) matrix of sub-band $\mathcal{B}^q$ is given by

$$\mathbf{R}(\mathcal{B}^q) = \mathbf{R}(\mathcal{A}_1) - \mathbf{R}(\mathcal{A}_2), \quad (32)$$

where $\mathcal{B}^q = \{f : f_{\min}^q \leq |f| \leq f_{\max}^q\}$, $\mathcal{A}_1 = \{f : |f| \leq f_{\max}^q\}$, and $\mathcal{A}_2 = \{f : |f| < f_{\min}^q\}$. Note that $\mathcal{B}^q = \mathcal{A}_1 - \mathcal{A}_2$. Since both $\mathbf{R}(\mathcal{A}_1)$ and $\mathbf{R}(\mathcal{A}_2)$ are Hermitian, their difference $\mathbf{R}(\mathcal{B}^q)$ is also Hermitian.

To show that $\mathbf{R}(\mathcal{B})$ is positive definite, note that $\mathcal{A}_2 \subset \mathcal{A}_1$. From Bronez's theorem [40], we have $\lambda_k(\mathcal{A}_2) < \lambda_k(\mathcal{A}_1)$, for $1 \leq k \leq K$, where $\lambda_k(\mathcal{A}_1)$ and $\lambda_k(\mathcal{A}_2)$ are the eigenvalues of the GPSS matrices for $\mathcal{A}_1$ and $\mathcal{A}_2$, respectively. Therefore,

$$\lambda_k(\mathcal{A}_1) - \lambda_k(\mathcal{A}_2) = \frac{\mathbf{x}_k^* \mathbf{R}(\mathcal{A}_1) \mathbf{x}_k}{\mathbf{x}_k^* \mathbf{R}(\mathcal{B}) \mathbf{x}_k} - \frac{\mathbf{x}_k^* \mathbf{R}(\mathcal{A}_2) \mathbf{x}_k}{\mathbf{x}_k^* \mathbf{R}(\mathcal{B}) \mathbf{x}_k}$$

$$= \frac{\mathbf{x}_k^* \mathbf{R}(\mathcal{B}^q) \mathbf{x}_k}{\mathbf{x}_k^* \mathbf{R}(\mathcal{B}) \mathbf{x}_k} > 0, \quad (33)$$

where the asterisk $*$ denotes complex conjugate transposition, and $\mathcal{B}$ is the entire signal band of any $\mathbf{x}$. Since $\mathbf{R}(\mathcal{B})$ in the denominator is positive definite, the numerator $\mathbf{x}_k^* \mathbf{R}(\mathcal{B}^q) \mathbf{x}_k$ must be positive. Thus, $\mathbf{R}(\mathcal{B}^q)$ is a positive definite Hermitian matrix.

B Sidelobe Leakage in Missing Data Sampling

Missing data sampling presents an interesting special case within the proposed framework. If the original sampling frequency is known, the optimal taper sequences (GPSS) for analysis bands $\mathcal{A}_0$ and $\mathcal{A}_i$, denoted as $\mathbf{w}_k^0$ and $\mathbf{w}_k^i$, can be related by a simple frequency shift operator E_i (20) [39, 40, 37]. Using the identity described in Section 3.5, the Generalized Eigenvalue Problem (GEP) equation for the nominal band $\mathcal{A}_0$ given by (18) is equivalent to

$$E_i^* \mathbf{R}(\mathcal{A}_i) E_i \mathbf{w}_k^0 = \lambda_k^0 \mathbf{R}(\mathcal{B}) \mathbf{w}_k^0. \quad (34)$$

Multiplying both sides by E_i to the left and using the property $E_i E_i^* = E_i^* E_i = \mathbb{I}$, we obtain

$$\mathbf{R}(\mathcal{A}_i) [E_i \mathbf{w}_k^0] = \lambda_k^0 [E_i \mathbf{R}(\mathcal{B}) E_i^*] [E_i \mathbf{w}_k^0]. \quad (35)$$

It can be shown [37] that for missing-data sampling with normalized sampling frequency 1 Hz and $f_{\max} = 0.5$ Hz, the GPSS matrix $\mathbf{R}(\mathcal{B})$ (9) becomes the identical matrix $\mathbb{I}$. Thus, $E_i \mathbf{R}(\mathcal{B}) E_i^* = \mathbf{R}(\mathcal{B})$ and the shifted tapers $\mathbf{w}_k^i = E_i \mathbf{w}_k^0$ satisfy the optimal criteria for bias (23) and variance (26). Moreover, since $\lambda_k^0 = \lambda_k^i$, sidelobe leakage $\mathcal{E}_i$ (29) is minimized (i.e., $\mathcal{E}_i = 0$).

However, if the original sampling frequency is incorrectly estimated, the sidelobe leakage is generally not optimized. For example, in the numerical experiments, the actual sampling rate was 1.2 Hz, but the normalized sampling frequency was assumed to be 1 Hz. This mismatch resulted in non-zero sidelobe leakage across the signal band. We formalize this situation in the following theorem:

Theorem 1. *For a missing data sampling scheme with an actual sampling frequency of $1/\beta$ Hz, where $\beta > 0$, normalized frequency of 1 Hz, and $f_{\max} = 1/2$ Hz, the sidelobe leakage estimate $\hat{L}_k^i = |\lambda_k^0 - \hat{L}_k^i|$ in (29) for the shifted tapers $\hat{\mathbf{w}}_k^i = E_i \mathbf{w}_k^0$ is bounded lower by*

$$\hat{L}_k^i \geq 0 \quad (36)$$

and upper by

$$\hat{L}_k^i \leq \lambda_k^0 \left(1 + \frac{f_w}{\|\mathbf{w}_k^0\|^2 - f_w} \right) < \infty, \text{ assuming } f_w < \|\mathbf{w}_k^0\|^2 \leq 2f_w. \quad (37)$$

Proof. To show the lower bound, we start with:

$$\begin{aligned} \hat{\lambda}_k^i &= \frac{(\hat{\mathbf{w}}_k^i)^* \mathbf{R}(\mathcal{A}_i) \hat{\mathbf{w}}_k^i}{(\hat{\mathbf{w}}_k^i)^* \mathbf{R}(\mathcal{B}) \hat{\mathbf{w}}_k^i} = \frac{\mathbf{w}_k^{0*} \mathbf{R}(\mathcal{A}_0) \mathbf{w}_k^0}{(E_i \mathbf{w}_k^0)^* \mathbf{R}(\mathcal{B}) (E_i \mathbf{w}_k^0)} \\ &= \frac{\lambda_k^0 \cdot 2f_w}{\mathbf{w}_k^{0*} \mathbf{E} \mathbf{R}(\mathcal{B}) \mathbf{w}_k^0}, \end{aligned} \quad (38)$$

where $\mathbf{E} \mathbf{R}(\mathcal{B}) \triangleq E_i^* \mathbf{R}(\mathcal{B}) E_i$, whose elements under the missing data sampling scheme are given by

$$\mathbf{E} \mathbf{R}(\mathcal{B}|n, m) = e^{-j2\pi f_{c_i} \beta k(n, m)} \mathbf{R}(\mathcal{B}|n, m) \quad (39)$$

and $k(n, n) = 0$, $|k(n, m)| \in \mathbb{Z}^+$ a positive integer for $n \neq m$, and $k(n, m) = -k(m, n)$, depending on a specific sampling scheme. If $\beta \in \mathbb{Z}$ is an integer, it is clear [37] that $\mathbf{ER}(\mathcal{B}) = \mathbf{R}(\mathcal{B}) = \mathbb{I}$ and thus $\hat{\lambda}_k^i = \lambda_k^0$, implying $\hat{L}_k^i = 0$.

To show the upper bound, let $P = \mathbf{w}_k^{0*} \mathbf{ER}(\mathcal{B}) \mathbf{w}_k^0 = \|\mathbf{w}_k^0\|^2 + S_f$. Here, $\|\mathbf{w}_k^0\|^2 = \sum_{n=1}^N \mathbf{w}_k^0(n)^* \mathbf{w}_k^0(n)$ is the sum of diagonal terms of P and S_f is the sum of off-diagonal terms given by

$$S_f = \sum_{n \neq m} \mathbf{ER}(\mathcal{B}|n, m) \mathbf{w}_k^{0*}(n) \mathbf{w}_k^0(m). \quad (40)$$

Since $P^* = P$, S_f is real-valued. Suppose $S \triangleq \sum_{n \neq m} \mathbf{R}(\mathcal{B}|n, m) \mathbf{w}_k^{0*} \mathbf{w}_k^0$, then S_f is the modulated version of S with phase rotation. From the normalization requirement (11), we have $|S| = 2f_w - \|\mathbf{w}_k^0\|^2$, assuming $\|\mathbf{w}_k^0\|^2 \leq 2f_w$.

In the worst-case destructive interference scenario, where the modulation of the off-diagonal terms to align in phase opposition to the diagonal contribution, minimizing the total value of P , we have $S_f \geq -|S| = \|\mathbf{w}_k^0\|^2 - 2f_w$. Therefore, assuming $f_w < \|\mathbf{w}_k^0\|^2$ and using triangle inequality, we have the upper bound of $\hat{L}_k^i$ given by

$$\begin{aligned} \hat{L}_k^i &\leq |\lambda_k^0| + |\hat{L}_k^i| = \lambda_k^0 \left(1 + \frac{2f_w}{|P|} \right) \\ &= \lambda_k^0 \left(1 + \frac{2f_w}{\|\mathbf{w}_k^0\|^2 + S_f} \right) \\ &\leq \lambda_k^0 \left(1 + \frac{2f_w}{\|\mathbf{w}_k^0\|^2 + (\|\mathbf{w}_k^0\|^2 - 2f_w)} \right) \\ &= \lambda_k^0 \left(1 + \frac{f_w}{\|\mathbf{w}_k^0\|^2 - f_w} \right) < \infty. \end{aligned} \quad (41)$$

□

This result shows that while frequency-shifted GPSS tapers can be optimal under ideal conditions, mismatches in sampling frequency introduce bounded but non-negligible sidelobe leakage. The assumption $f_w < \|\mathbf{w}_k^0\|^2 \leq 2f_w$ is typical since $\mathbf{R}(\mathcal{B}|n, m) = 1$ for $n = m$, and $\mathbf{R}(\mathcal{B})$ is positive definite. Finally, we note that the tapers $\mathbf{w}_k^0$ are optimal GPSS for the nominal band $\mathcal{A}_0$, but may not be equivalent to Missing-Data Slepian Sequence (MDSS) used in [37].

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